

EXPERIMENTAL STUDY OF PARALLELLY CONNECTED SEIGs WITH 4- WIRE SYSTEM USING HYBRID EXITATION FOR RURAL AREA

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Thesis

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By

Mr. Deepak Kumar (EE/20/17)

Mr. Bhupendra Singh Senger (EE/20/12)

Kundan Meena (EE/20/15)

Madap Waghe (EE/20/16)

Saurav Kumar Yadav (EE/20/14)

Under the supervision of:

Dr. Rajen Pudur

Associate Professor

Department of Electrical Engineering



DEPARTMENT OF ELECTRICAL ENGINEERING

National Institute of Technology Arunachal Pradesh

(Established by Ministry of Education, Govt. of India)

Jote, District: Papum Pare, Arunachal Pradesh 791 113

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National Institute of Technology, Arunachal Pradesh

(Institute of National Importance under Ministry of Education, Govt. of India)

Jote – 791113, Dist.: Papum Pare, Arunachal Pradesh

(E-Mail nitarunachal@nitap.ac.in/registrarcell@nitap.ac.in)

Website: www.nitap.ac.in,

Ph: 0360-295454

CERTIFICATE OF APPROVAL

The dissertation entitled experimental study of parallelly connected SEIGs with 4- wire system using hybrid excitation for rural area submitted by Deepak Kumar, Bhupendra Singh Senger, Kundan Meena, Madap Waghe, Saurav Kumar Yadav being Registration No. 0000001333/A/2021, 0000001328/A/2021, 0000001331/A/2021, 0000001324/A/2021, 0000001330/A/2021 is presented in a satisfactory manner to warrant its acceptance as a prerequisite for the degree of Bachelor of Technology in Electrical Engineering of the National Institute of Technology, Arunachal Pradesh. It is understood that by this approval the undersigned do not necessarily endorse or approve any statement made, opinion expressed or conclusion drawn therein, but only for the purpose for which it has been submitted.

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National Institute of Technology, Arunachal Pradesh

(Institute of National Importance under Ministry of Education, Govt. of India)

Jote – 791113, Dist.: Papum Pare, Arunachal Pradesh

(E-Mail nitarunachal@nitap.ac.in/registrarcell@nitap.ac.in)

Website: www.nitap.ac.in,

Ph: 0360-295454

CERTIFICATE FROM SUPERVISOR

This is to certify that the dissertation entitled experimental study of parallelly connected SEIGs with 4- wire system using hybrid excitation for rural area submitted by Deepak Kumar, Bhupendra Singh Senger ,Kundan Meena, Madap Waghe and Saurav Kumar Yadav being Registration No. 0000001333/A/2021, 0000001328/A/2021, 0000001331/A/2021, 0000001324/A/2021 and 0000001330/A/2021 to the Department of Electrical Engineering of the National Institute of Technology, Arunachal Pradesh, as a partial fulfilment of their B. Tech in Electrical Engineering of the Institute is absolutely based upon their own work, carried out during the period from January – May, 2024 under my supervision. Neither this dissertation nor any part of it has been submitted for the award of any other degree of this Institute or any other Institute/University.

(Dr. Rajen Pudur)

Supervisor

Date: 17/05/24

(Official Seal)



National Institute of Technology, Arunachal Pradesh
(Institute of National Importance under Ministry of Education, Govt. of India)
Jote – 791113, Dist.: Papum Pare, Arunachal Pradesh

(E-Mail nitarunachal@nitap.ac.in/registrarcell@nitap.ac.in)

Website: www.nitap.ac.in,

Ph: 0360-295454

PLAGIARISM UNDERTAKING CERTIFICATE

This is to certify that the dissertation entitled **“EXPERIMENTAL STUDY OF PARALLELLY CONNECTED SEIGs WITH 4- WIRE SYSTEM USING HYBRID EXITATION FOR RURAL AREA”** submitted by **Mr. Deepak Kumar, Mr. Bhupendra Singh Senger , Mr. Kundan Meena, Mr. Madap Waghe and Mr. Saurav Kumar Yadav** bearing Registration No. **0000001333/A/2021, 0000001328/A/2021, 0000001331/A/2021, 0000001324/A/2021 and 0000001330/A/2021** to the Department of **Electrical Engineering** of the National Institute of Technology, Arunachal Pradesh, as a partial fulfilment of his B. Tech Degree in Electrical Engineering of the Institute is free from any plagiarism articles with similarity index of 12%. Each chapter is an outcome of independent and original work; thus, duly acknowledge all sources from which the ideas and extracts have been taken in adherence to the law of anti-plagiarism.

Deepak Kumar
Bhupendra Singh Senger
Kundan Meena
Madap Waghe
Saurav Kumar Yadav
Date: 17/05/24

(Dr. Rajen Pudur)
Supervisor
Date: 17/05/24
(Official Seal)

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Mr. Deepak Kumar (EE/20/17)

Mr. Bhupendra Singh Senger (EE/20/12)

Kundan Meena (EE/20/15)

Madap Waghe (EE/20/16)

Saurav Kumar Yadav (EE/20/14)

Place: Jote, Papum Pare

Dated 17.05.2024

ABSTRACT

This thesis presents a comprehensive study on the steady-state modelling, analysis, and experimental investigation of self-excited induction generators (SEIGs) for distributed generation and renewable energy systems in rural electrification applications. The study focuses on the development and evaluation of a hybrid excitation system for parallel-connected SEIGs with a four-wire configuration. A novel approach based on coordinate geometry principles is proposed for the steady-state modelling of SEIGs, simplifying the analysis by representing the SEIG's equivalent circuit admittances as points in the complex plane. A closed-form expression for the per-unit speed and magnetizing reactance is derived, enabling accurate prediction of the SEIG's performance characteristics. Experimental analysis is conducted to identify the optimal winding configuration for SEIGs in a four-wire system, considering various combinations of machine windings, capacitor banks, and load connections. The delta-star-star combination emerges as the most promising configuration, exhibiting superior voltage regulation, power output, and overall system efficiency. The parallel operation of two micro-hydro power plants (2.2 kW and 5.5 kW) employing a hybrid excitation system is investigated. The hybrid excitation technique combines star and delta configurations of the capacitor banks with a neutral connection. Experimental results demonstrate the effectiveness of the hybrid excitation system in maintaining load balance, voltage regulation, and power quality under various loading conditions, including resistive, inductive, and non-linear loads. The performance of the hybrid excitation system under unbalanced load conditions is also studied, revealing its robustness and adaptability in maintaining a stable frequency despite increasing load imbalances. However, voltage imbalance and neutral currents increase with higher load imbalances, emphasizing the importance of proper system design and control strategies. The successful parallel operation of SEIGs with the hybrid excitation system under diverse load conditions highlights its potential for distributed generation and renewable energy applications. The system demonstrates stable voltage and frequency characteristics, effective reactive power compensation, and near-unity power factor operation. The experiment outcomes provide valuable insights into the steady-state modelling, optimal winding configuration, and parallel operation of SEIGs with a hybrid excitation system. The proposed coordinate geometry approach simplifies the analysis and offers a computationally efficient alternative to existing techniques. The experimental results validate the effectiveness of the hybrid excitation system in enhancing the performance and stability of parallel-connected SEIGs, even under challenging load conditions.

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List of notations

Notations	Descriptions
P_e	Electrical Power, Watt
IG	Induction Generator
HP	Horse Power
Q_g	Reactive Power
V	Voltage
F	Frequency
C	Capacitance
μf	Micro Farad
DC	Direct Current
Hz	Frequency
kW	Kilowatt
IM	Induction Machine
Z_a	Impedance of Phase A
Z_b	Impedance of Phase B
Z_c	Impedance of Phase C
R	Resistance
I	Current
AI	Artificial Intelligence
IoT	Internet of Things

List of nomenclature

Variable	Description
a	Per unit (pu) frequency, defined as the ratio of the generated frequency (f_g) to the rated frequency (f_r)
b	Per unit (pu) speed, defined as the ratio of the actual rotor speed (N_r) to the synchronous speed at rated frequency (N_s)
X_m	Per phase magnetizing reactance (Ω)
C	Per phase excitation capacitance (F)
E	Per phase air gap voltage (V)
f_g	Generated frequency (Hz)
f_r	Rated frequency (Hz)
N_r	Actual speed of the rotor
N_s	Synchronous speed at rated frequency (rpm)
R	Per phase load resistance (Ω)
X	Per phase load reactance (Ω)
R_1	Per phase stator resistance (Ω)
R_2	Per phase rotor resistance (referred to the stator) (Ω)
X_1	Per phase stator reactance (Ω)
X_2	Per phase rotor reactance (referred to the stator) (Ω)
X_C	Per phase excitation capacitance reactance (Ω)
X_{mc}	Per phase critical magnetizing reactance (Ω)
Y	Net admittance of the SEIG equivalent circuit
Y_1	Admittance of the stator branch
Y_2	Admittance of the magnetizing branch

Y_3	Admittance of the rotor branch
Y_{1r}	Real part of the admittance Y_1
Y_{1i}	Imaginary part of the admittance Y_1
Y_{2r}	Real part of the admittance Y_2
Y_{2i}	Imaginary part of the admittance Y_2
Y_{3i}	Imaginary part of the admittance Y_3
Z	Load impedance
Z_C	Excitation capacitance impedance
Z_L	Net impedance of the parallel combination of load impedance (Z) and excitation capacitance impedance (Z_C)
R_L	Real part of the net impedance Z_L
X_L	Imaginary part of the net impedance Z_L

List of acronyms

SEIG	SEIG: Self-Excited Induction Generator
DSS	Delta-machine, Star excitation capacitor arrangement and load
VFD	Variable Frequency Drive
THD	Total Harmonic Distortion
RMS	Root Mean Square

CHAPTER 1

Introduction of SEIG

CHAPTER 1

Introduction of SEIG

1.1 Introduction

The pursuit of sustainable and renewable energy sources has gained significant momentum in recent years, driven by the pressing need to address global environmental challenges and energy security concerns. In this context, the self-excited induction generator (SEIG) has emerged as a promising solution for decentralized power generation, particularly in remote and off-grid areas where traditional grid infrastructure is lacking or uneconomical to establish. SEIGs are derived from conventional induction motors, which are widely available and cost-effective. By connecting a suitable capacitor bank across the stator terminals of an induction motor, the machine can be transformed into a self-excited induction generator capable of producing electrical power. This simple yet ingenious concept was first introduced by Basset and Potter in 1935, laying the foundation for the development of SEIGs as a viable alternative for small-scale power generation. The principle of self-excitation in SEIGs relies on the residual magnetism present in the rotor circuit or the magnetizing current drawn from the capacitor bank during start-up. This initial magnetizing current establishes a rotating magnetic field in the air gap, which induces voltages in the stator windings. When the load current exceeds specific limitations, the residual magnetism decreases to zero, and the machine is de-excited. When this occurs, there are four options for re-magnetization.

1. High value capacitor is used in place of excitation capacitor to regenerate the residual magnetism.
2. A charged battery can also support to restore the residual magnetism.
3. AC to DC converter can also be used.
4. By running the SEIG above the normal speed for about 10 minutes also helps to regain the residual magnetism.

These induced voltages, in turn, create currents that flow through the capacitor bank, providing the necessary reactive power for sustained self-excitation. Once the self-excitation process is initiated, the SEIG can operate independently, generating electrical power as long as a prime mover (such as a hydro turbine, wind turbine, or diesel engine) drives the rotor. One of the key advantages of SEIGs is their ability to operate in isolated mode, without the need for an external source of excitation or a synchronous link to the grid. This feature makes them particularly

suitable for remote and off-grid applications, where the extension of traditional power transmission lines is often impractical or economically unfeasible.

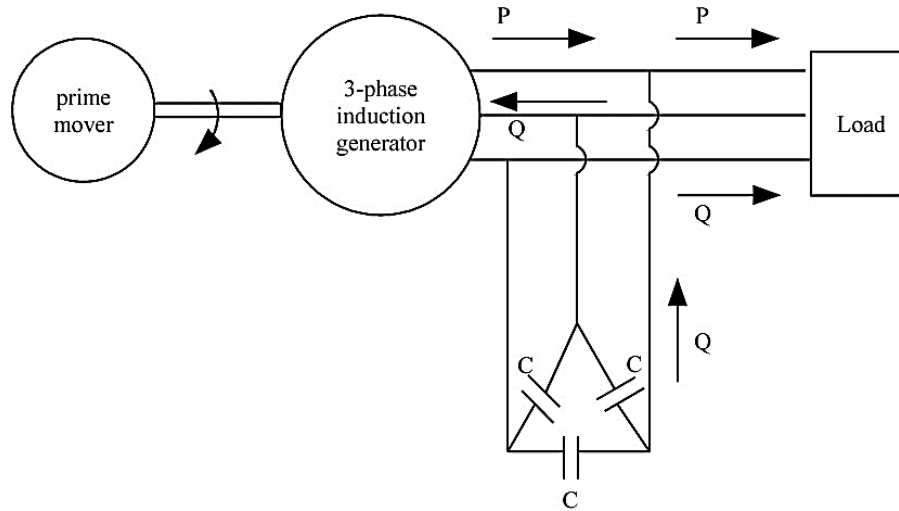


Fig. 1.1: Schematic diagram of SEIG with excitation and load

Despite their simplicity and cost-effectiveness, SEIGs face several challenges that have been the subject of extensive research and development efforts. One of the primary challenges is the voltage regulation and power quality, which can be influenced by various factors such as load variations, capacitor bank configuration, and machine parameters.

To address the issue of voltage regulation and power quality, researchers have explored various techniques, including the implementation of a four-wire system and the parallel operation of multiple SEIGs with hybrid excitation. The four-wire system, which is one of the primary objectives of this study, involves the addition of a neutral wire to the conventional three-wire system. This configuration allows for the connection of single-phase loads, enhancing the versatility and applicability of SEIGs in residential and small-scale industrial settings.

The analysis and evaluation of different winding configurations for the SEIG, capacitor bank, and load connections in the four-wire system is a crucial aspect of this research. By systematically investigating various combinations of delta and star connections, the study aims to identify the optimal arrangement that offers superior voltage regulation, power quality, and overall system efficiency.

Furthermore, the study explores the parallel operation of two micro-hydro power plants, each equipped with a SEIG of different capacity (2.2 kW and 5.5 kW), employing a hybrid excitation technique. The hybrid excitation approach involves combining both star and delta configurations of the capacitor bank with a neutral connection. This innovative technique has

shown promising results in terms of improved voltage regulation and load-sharing capabilities, making it an attractive solution for scaling up and integrating multiple SEIGs into a unified power generation system.

The motivation behind this research is multifaceted and driven by the pressing need to address the energy challenges faced by remote and underserved communities. Access to reliable and affordable electricity is a crucial factor in promoting socio-economic development, improving quality of life, and fostering sustainable growth. By exploring innovative solutions such as four-wire SEIG systems and parallel operation with hybrid excitation, this study aims to contribute to the advancement of decentralized power generation technologies, enabling the deployment of cost-effective and efficient micro-hydro power plants. Moreover, the integration of multiple micro-hydro power plants through parallel operation addresses the scalability and reliability concerns associated with standalone systems. By combining the output of multiple SEIGs, the overall power generation capacity can be increased to meet the growing energy demands of remote communities. Additionally, the parallel operation provides redundancy, enhancing the system's resilience and reducing the impact of individual unit failures or maintenance requirements.

Beyond the technical aspects, this thesis aligns with the global efforts towards sustainable development and environmental conservation. The utilization of renewable energy sources, such as micro-hydro, in conjunction with efficient and reliable power generation technologies like SEIGs, contributes to the reduction of greenhouse gas emissions and the mitigation of climate change impacts. By promoting the widespread adoption of these solutions, particularly in remote and off-grid areas, this study supports the transition towards a more sustainable and environmentally friendly energy landscape.

Furthermore, the successful implementation of four-wire SEIG systems and parallel operation with hybrid excitation can have far-reaching implications for rural electrification projects and the development of decentralized energy infrastructure. By overcoming the limitations of traditional power transmission lines and centralized generation, these solutions empower communities to harness local renewable resources and achieve energy self-sufficiency, fostering economic growth and improving overall quality of life.

This thesis aims to address the challenges associated with voltage regulation, power quality, and scalability in self-excited induction generator systems. By investigating the four-wire system and parallel operation with hybrid excitation, the study seeks to contribute to the

development of innovative and sustainable solutions for decentralized power generation. The findings and insights gained from this research have the potential to catalyze the widespread adoption of SEIGs, particularly in remote and off-grid areas, supporting global efforts towards sustainable development and energy access for all.

1.2 Motivation

The self-excited induction generator (SEIG) has gained significant attention in recent years due to its potential applications in renewable energy systems, particularly in remote areas and developing countries. The SEIG offers a cost-effective and reliable solution for generating electricity from renewable sources, such as micro-hydro, wind, or biomass. However, one of the challenges associated with SEIGs is the need for a balanced three-phase load, which is often not readily available in rural or remote areas. To address this issue, the implementation of a four-wire system with SEIGs has emerged as a promising approach. By incorporating a neutral wire, the four-wire system allows for the connection of single-phase loads, making it more suitable for residential and small-scale industrial applications. This configuration not only enhances the versatility of SEIGs but also promotes efficient energy utilization and improved power distribution. The primary objective of this study is to investigate the self-excited induction generator with a four-wire system, focusing on the analysis and evaluation of different wiring configurations. By employing two SEIGs of different capacities (2.2 kW and 5.5 kW), the research aims to determine the most effective combination of winding configurations, including delta and star connections for the machine, capacitor bank, and load. This comprehensive analysis will provide valuable insights into the optimal setup for maximizing the performance and efficiency of the four-wire SEIG system.

Furthermore, the study aims to explore the parallel operation of two micro-hydro power plants (2.2 kW and 5.5 kW) utilizing a hybrid excitation technique. The hybrid excitation approach combines both star and delta configurations of the capacitor bank with a neutral connection, offering potential benefits in terms of load management, voltage regulation, and power quality. By investigating this parallel operation, the research seeks to develop strategies for integrating multiple micro-hydro power plants into a unified system, enhancing the overall reliability and scalability of the renewable energy infrastructure. The motivation behind this study is multifaceted. Firstly, it addresses the pressing need for accessible and reliable electricity in remote and underserved areas, where traditional grid infrastructure may be lacking or inadequate. By leveraging the potential of SEIGs and exploring the four-wire system, the

research aims to contribute to the development of decentralized, sustainable energy solutions that can improve the quality of life for communities in these regions. Secondly, the study holds significant importance for the advancement of renewable energy technologies. The findings and insights gained from this research can pave the way for more efficient and cost-effective implementations of micro-hydro and other renewable energy systems, thereby promoting the transition towards a more sustainable and environmentally friendly energy landscape. Moreover, the parallel operation aspect of the study addresses the need for scalability and integration of multiple power sources, which is crucial for meeting the growing energy demands of rural and remote communities. By developing strategies for parallel operation, the research aims to enhance the reliability and resilience of the overall energy system, ensuring a consistent and uninterrupted supply of electricity.

This Thesis represents a crucial step towards realizing the full potential of self-excited induction generators and micro-hydro power plants, while simultaneously contributing to the global efforts towards sustainable development and environmental conservation. The motivation driving this research is rooted in the pursuit of innovative solutions that can bridge the gap between energy needs and available resources, empowering communities and promoting socio-economic growth responsibly and sustainably.

1.3 Objectives

1. To conduct a comprehensive study on the self-excited induction generator (SEIG) with a four-wire system, investigating its potential applications and operational characteristics.
2. To systematically analyse and evaluate various winding configurations for the SEIG, including delta and star connections for the machine windings, capacitor bank, and load connections, with the aim of determining the most efficient and optimal combination for the four-wire system.
3. To explore the parallel operation of two micro-hydro power plants (2.2 kW and 5.5 kW) employing a hybrid excitation technique, which combines both star and delta configurations of the capacitor bank with a neutral connection, and assess its performance in terms of load management, voltage regulation, and power quality.

1.4 Literature Review

The self-excited induction generator (SEIG) has emerged as a viable solution for generating electricity from renewable sources, particularly in remote areas and developing countries. This

literature review aims to provide a comprehensive overview of the existing research and studies related to SEIGs, with a particular focus on the four-wire system and parallel operation with hybrid excitation. The concept of self-excitation in induction generators was first introduced by Basset and Potter in 1935 [1]. They demonstrated that an induction machine could operate as a generator by connecting a suitable capacitor bank across its terminals, providing the necessary excitation. Since then, numerous researchers have explored various aspects of SEIGs, including their steady-state and transient performance, voltage regulation, and protection schemes [2-3]. One of the key challenges associated with SEIGs is the need for a balanced three-phase load, which is often not readily available in remote or rural areas. To address this issue, the implementation of a four-wire system with SEIGs has been proposed by several researchers [4-5]. The four-wire system allows for the connection of single-phase loads, making it more suitable for residential and small-scale industrial applications. Rahim et al. [6] investigated the performance of a four-wire SEIG system and concluded that it can effectively supply both single-phase and three-phase loads while maintaining acceptable voltage regulation. The winding configurations of the SEIG, capacitor bank and load connections play a crucial role in the overall performance and efficiency of the system. Several studies have explored different combinations of delta and star connections for the machine windings, capacitor banks, and load connections [7-8]. Murthy et al. [9] analysed the performance of a four-wire SEIG system with various winding configurations and found that the delta-connected SEIG with a star-connected capacitor bank and star-connected load exhibited the best voltage regulation and efficiency. In addition to the winding configurations, the capacitance value of the excitation capacitor bank is another critical factor that affects the performance of SEIGs. Researchers have proposed various analytical methods and optimization techniques to determine the optimal capacitance value for different operating conditions and load scenarios [10-11]. The parallel operation of multiple SEIGs has gained significant attention due to its potential for improving the reliability and scalability of renewable energy systems. Murthy et al. [12] investigated the parallel operation of two SEIGs and proposed a control strategy for voltage regulation and load sharing. Singh et al. [13] studied the parallel operation of three SEIGs and developed a mathematical model to analyse the steady-state performance. To further enhance the performance of parallel SEIG systems, researchers have explored the concept of hybrid excitation, which involves combining both star and delta configurations of the capacitor bank with a neutral connection [14-15]. Mrinal et. al. [16] introduces a novel technique to parallel two SEIG. Ojo et al. [17] investigated the hybrid excitation approach for a parallel operation of two SEIGs and reported improved voltage regulation and load-sharing

capabilities. Several studies have focused on the application of SEIGs in micro-hydro power plants, which are particularly suitable for remote and off-grid communities [18-19]. In addition to experimental and analytical studies, researchers have also developed computational models and simulations to analyse the performance of SEIGs under various operating conditions and configurations [21-22]. These computational tools have proven valuable in designing and optimizing SEIG systems for specific applications. While significant progress has been made in the field of SEIGs, there are still several areas that require further research and investigation. These include advanced control strategies for voltage regulation and load sharing in parallel SEIG systems, fault analysis and protection schemes, and the integration of SEIGs with other renewable energy sources and energy storage systems [23-24].

1.5 Organization of the Thesis

The thesis is organized into eight chapters, each focusing on a specific aspect of the study on self-excited induction generators (SEIGs) and their applications in rural electrification. Chapter 1 introduces SEIGs, discussing their potential for decentralized power generation, the research motivation, objectives, and a comprehensive literature review. Chapter 2 presents the steady-state mathematical modelling of SEIGs with different capacitor and load configurations, including star-star-star, delta-delta-star, delta-star-star, and star-delta-star combinations. Chapter 3 introduces a hybrid excitation topology combining star and delta connections to address off-grid application requirements, comparing it with the conventional DSS topology. Chapter 4 describes the experimental setup, investigates various SEIG configurations, and calculates SEIG parameters through tests such as blocked rotor, no load, and magnetic saturation. Chapter 5 focuses on the hybrid excitation-based parallel operation of SEIGs with different power ratings for rural applications, analysing their performance under resistive, inductive-resistive, and nonlinear loads. Chapter 6 discusses the results of the hybrid excitation approach and parallel operation of SEIGs under various load conditions, providing insights into system performance and efficiency. Chapter 7 summarizes the findings and potential applications of the hybrid excitation system for parallel-connected SEIGs in rural electrification projects. Finally, Chapter 8 outlines areas for future research and development related to hybrid excitation and SEIG systems, identifying potential avenues for further exploration and improvement.

CHAPTER 2

Mathematical modelling of the SEIG with different capacitor and load configurations

CHAPTER 2

Mathematical modelling of the SEIG with different capacitor and load configurations

2.1 Introduction

A self-excited induction generator (SEIG) is an induction machine that operates as a generator without requiring an external power supply for excitation. Instead, it relies on the residual magnetism in the rotor and the presence of capacitors connected to the stator windings to build up and maintain the voltage across the generator terminals.

The performance of an SEIG depends on various factors, including the machine parameters, capacitor size, and load characteristics. The choice of capacitor and load combination significantly influences the voltage regulation, frequency stability, and overall efficiency of the generator. In this chapter, discuss four common capacitor and load combinations for SEIGs: star-star-star, delta-delta-star, delta-star-star, and star-delta-star and also present the steady-state mathematical modelling for each combination.

1. **Star-Star-Star Combination:** In this configuration, the machine windings, capacitors, and load are all connected in a star (wye) configuration. The star connection provides a neutral point, which can be grounded for safety purposes. The steady-state mathematical modelling for this combination involves writing the voltage and current equations for each phase, considering the machine parameters, capacitor values, and load impedance.

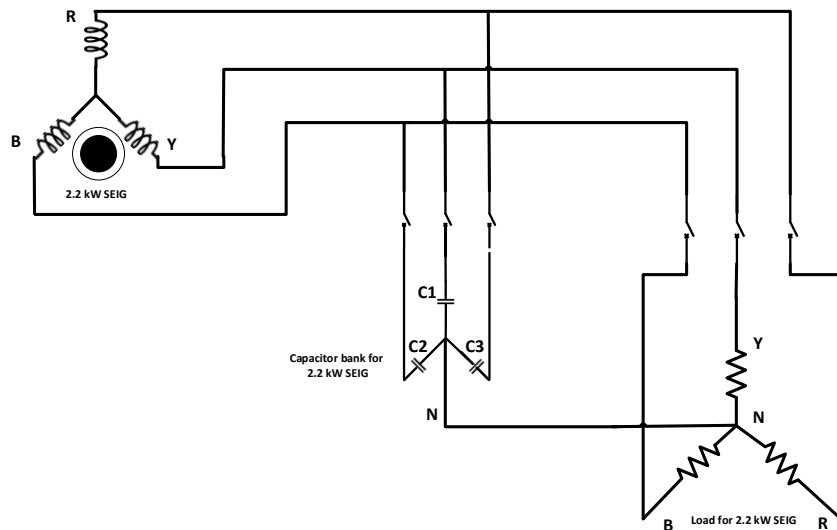


Fig. 2.1: Star-Star-Star Combination

2. **Delta-Delta-Star Combination:** This combination features the machine windings and capacitors connected in a delta configuration, while the load is connected in a star configuration. The delta connection eliminates the need for a neutral point and offers higher voltage and current ratings compared to the star connection. The steady-state mathematical modelling for this combination involves transforming the delta-connected parameters into their equivalent star-connected values and then writing the voltage and current equations for each phase.

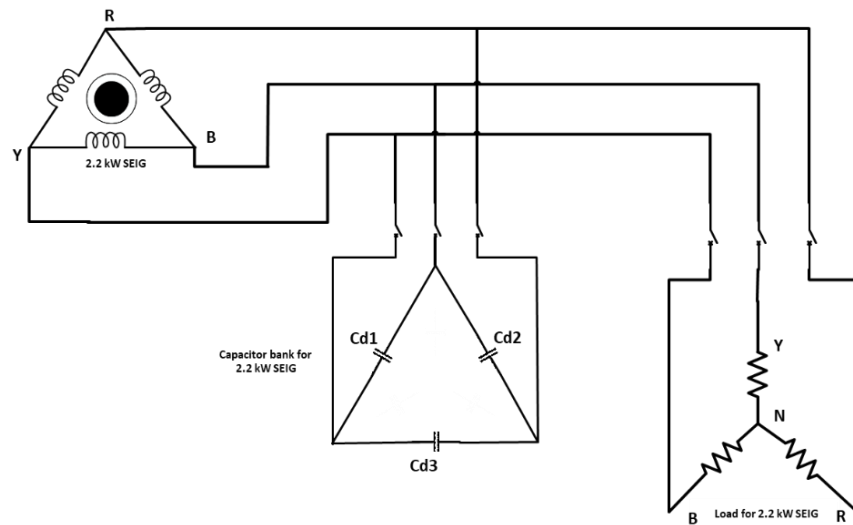


Fig. 2.2: Delta-Delta-Star Combination

3. **Delta-Star-Star Combination:** In this configuration, the machine windings are connected in a delta configuration, while the capacitors and load are connected in a star configuration.

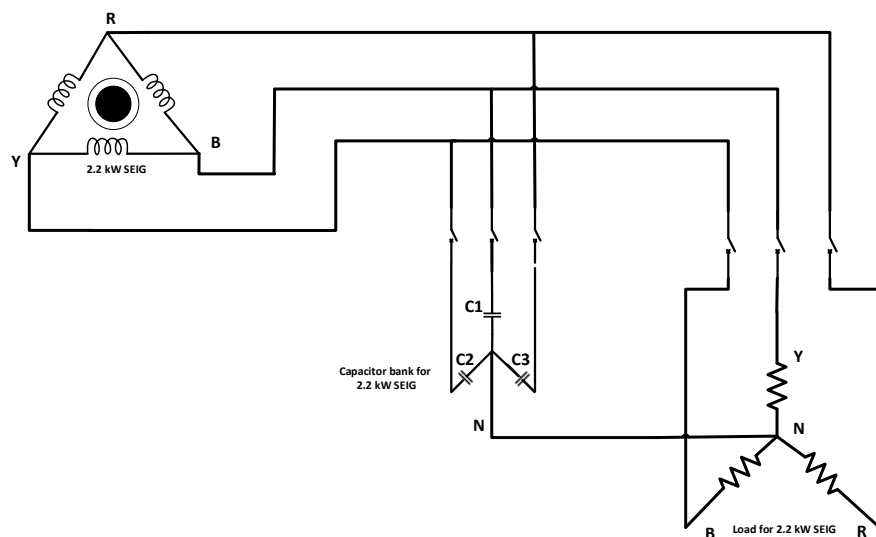


Fig. 2.3: Delta-Star-Star Combination

This combination provides a balance between the advantages of delta and star connections, such as improved voltage regulation and reduced harmonic distortion. The steady-state mathematical modelling for this combination involves transforming the delta-connected machine parameters into their equivalent star-connected values and then writing the voltage and current equations for each phase, considering the star-connected capacitors and load.

4. **Star-Delta-Star Combination:** This combination features the machine windings connected in a star configuration, while the capacitors are connected in a delta configuration, and the load is connected in a star configuration. The star connection of the machine windings provides a neutral point, while the delta connection of the capacitors offers higher voltage and current ratings. The steady-state mathematical modelling for this combination involves writing the voltage and current equations for each phase, considering the star-connected machine parameters, delta-connected capacitor values, and star-connected load impedance.

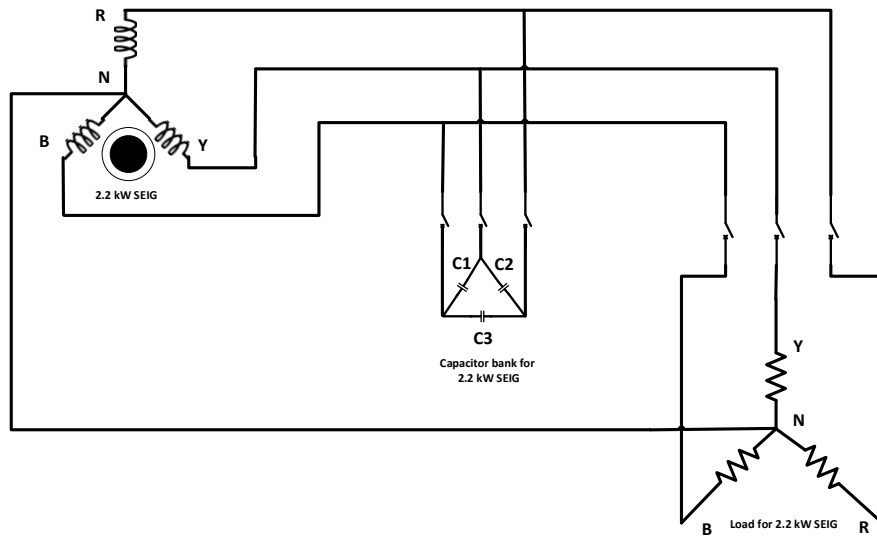


Fig. 2.4: Star-Delta-Star Combination

The steady-state mathematical modelling for each combination involves writing the voltage and current equations based on Kirchhoff's laws and the machine's equivalent circuit. The equations consider the machine parameters, such as stator and rotor resistances, leakage and magnetizing inductances, and the capacitor and load values. By solving these equations, various performance characteristics of the SEIG, such as voltage magnitude, frequency, power output, and efficiency, can be determined for each capacitor and load combination.

2.2 Steady-state mathematical modelling.

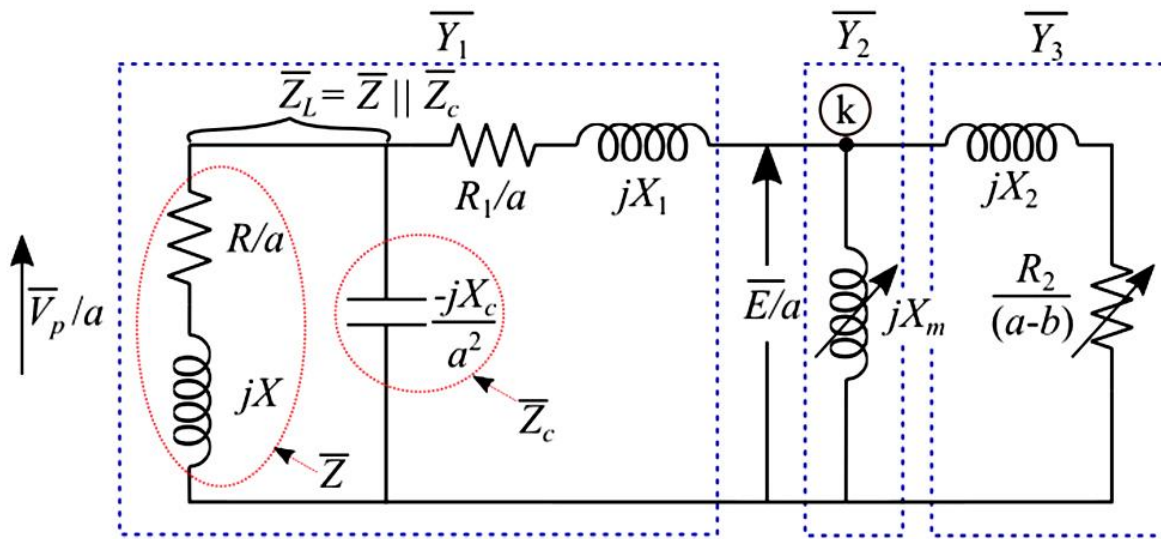


Fig. 2.5: steady-state Equivalent circuit of SEIG

The steady-state equivalent circuit of the self-excited induction generator (SEIG) is depicted in Figure 2.5. This circuit representation considers the variation of the magnetizing reactance, which is influenced by the level of magnetic saturation in the machine. The magnetizing reactance plays a crucial role in the voltage build-up process and the overall performance of the SEIG. It is important to note that the equivalent circuit neglects the core loss, which simplifies the analysis. Additionally, all the reactance values shown in the equivalent circuit correspond to the rated frequency of the induction machine, which is a key parameter in the design and operation of the SEIG. The equivalent circuit provides a convenient way to analyse the steady-state behaviour of the SEIG and predict its performance under various operating conditions.

The net impedance Z_L is obtained by considering the parallel combination of the load impedance Z and the excitation capacitance impedance Z_c .

$$Z_L = \frac{Z \times Z_c}{Z + Z_c} \quad (1)$$

Where $Z = \frac{R}{a} + jX$ and $Z_c = \frac{-jX_c}{a^2}$

Substituting Z and Z_c into (1)

$$Z_L = \frac{\left(\frac{R}{a} + jX\right) \times \left(-j\frac{X_c}{a^2}\right)}{\left(\frac{R}{a} + jX\right) + \left(-j\frac{X_c}{a^2}\right)}$$

Now simplifying the equation Z_L .

$$Z_L = \frac{(-j \frac{RX_c}{a^3} - j \frac{XX_c}{a^2})}{(\frac{R}{a} - j \frac{X_c}{a^2}) + (jX)}$$

Multiplying the numerator and denominator by the complex conjugate of the denominator.

$$Z_L = \frac{(-j \frac{RX_c}{a^3} - j \frac{XX_c}{a^2}) \times (\frac{R}{a} + j \frac{X_c}{a^2} - jX)}{\left(\left(\frac{R}{a}\right)^2 + \left(\frac{X_c}{a^2}\right)^2\right) + X^2}$$

Expanding the numerator

$$Z_L = \frac{(\frac{RX_c^2}{a^5} + j \frac{RXX_c}{a^3} - j \frac{RXX_c}{a^3} + \frac{XX_c^2}{a^4})}{\left(\left(\frac{R}{a}\right)^2 + \left(\frac{X_c}{a^2}\right)^2\right) + X^2}$$

Simplifying

$$Z_L = \frac{(\frac{RX_c^2}{a^5} + \frac{XX_c^2}{a^4})}{\left(\left(\frac{R}{a}\right)^2 + \left(\frac{X_c}{a^2}\right)^2\right) + X^2} + j \frac{(\frac{RXX_c}{a^3} - \frac{RXX_c}{a^3})}{\left(\left(\frac{R}{a}\right)^2 + \left(\frac{X_c}{a^2}\right)^2\right) + X^2}$$

The real and imaginary parts of Z_L be evaluated as

$$R_L = \text{real}(Z_L) = \frac{(\frac{RX_c^2}{a^5} + \frac{XX_c^2}{a^4})}{\left(\left(\frac{R}{a}\right)^2 + \left(\frac{X_c}{a^2}\right)^2\right) + X^2}$$

(2)

$$X_L = \text{imag}(Z_L) = \frac{(\frac{RXX_c}{a^3} - \frac{RXX_c}{a^3})}{\left(\left(\frac{R}{a}\right)^2 + \left(\frac{X_c}{a^2}\right)^2\right) + X^2} = 0$$

For SEIG operation, the net reactance part X_L should be capacitive to overcome the lagging reactive power drawn by inductive reactance.

Now determine the net admittance Y .

The net admittance Y , consisting of Y_1 , Y_2 and Y_3 in parallel, is given by:

$$Y = Y_1 + Y_2 + Y_3 \quad (3)$$

Where $Y_1 = \frac{1}{\left(\frac{R_1}{a} + R_L + j(X_1 - X_L)\right)}$, $Y_2 = \frac{1}{jX_m}$ and $Y_3 = \frac{1}{\left(\frac{R_2}{(a-b)} + jX_2\right)}$

Substituting the expressions for Y_1 , Y_2 and Y_3 into (3).

$$Y = \frac{1}{\left(\frac{R_1}{a} + R_L + j(X_1 - X_L)\right)} + \frac{1}{jX_m} + \frac{1}{\left(\frac{R_2}{(a-b)} + jX_2\right)}$$

Now applying nodal analysis.

There is no external voltage or current source in the circuit, and the induced EMF E is produced by self-excitation only. Applying nodal analysis at node k , the nodal equation can be written as.

$$Y \times \frac{E}{a} = 0 \quad (4)$$

Under steady-state operation, there should be a finite induced EMF E produced by the SEIG. Therefore, E cannot be zero, and Y has to be zero in (4).

Separate real and imaginary parts of Y by Applying the concept from (3) can be written as follows, wherein each term is separated into real and imaginary parts.

$$(Y_{1r} + jY_{1i}) + (Y_{2r} + jY_{2i}) + (Y_{3r} + jY_{3i}) = 0 \quad (5)$$

Where

$$Y_1 = \frac{1}{\left(\frac{R_1}{a} + R_L + j(X_1 - X_L)\right)}$$

$$Y_{1r} = \frac{\left(\frac{R_1}{a} + R_L\right)}{\left(\frac{R_1}{a} + R_L\right)^2 + (X_1 - X_L)^2} \quad (6)$$

$$Y_{1i} = -\frac{X_1 - X_L}{\left(\frac{R_1}{a} + R_L\right)^2 + (X_1 - X_L)^2} \quad (7)$$

$$Y_2 = \frac{1}{jX_m}$$

$$Y_{2r} = 0 \quad (8)$$

$$Y_{2i} = -\frac{1}{X_m} \quad (9)$$

$$Y_3 = \frac{1}{\left(\frac{R_2}{(a-b)} + jX_2\right)}$$

$$Y_{3r} = \frac{\frac{R_2}{(a-b)}}{\left(\frac{R_2}{(a-b)}\right)^2 + X_2^2} \quad (10)$$

$$Y_{3i} = -\frac{X_2}{\left(\frac{R_2}{(a-b)}\right)^2 + X_2^2} \quad (11)$$

The equations (6) to (11) represent the real and imaginary parts of the admittances Y_1 , Y_2 and Y_3 .

The equations (6) to (11) represent the real and imaginary parts of the admittances Y_1 , Y_2 and Y_3 . These equations form a system of nonlinear equations that need to be solved simultaneously to determine the unknown parameters and steady-state performance of the SEIG. There are several methods to solve these equations. In our work, we used the Newton-Raphson Method to solve using MATLAB software. the algorithm and MATLAB code are shown in Appendix 1.

First, defining the system of equations as

$$\begin{aligned} f_1(a, b, X_m) &= Y_{1r} + Y_{2r} + Y_{3r} = 0 \\ f_2(a, b, X_m) &= Y_{1i} + Y_{2i} + Y_{3i} = 0 \end{aligned}$$

By choosing initial guesses for the unknown variables a , b , and X_m . Let's denote them as a_0 , b_0 , and X_{m0} . The Jacobian matrix J is the matrix of partial derivatives of the functions f_1 and f_2 with respect to the unknown variables a , b , and X_m .

$$J = \begin{bmatrix} \frac{\partial f_1}{\partial a} & \frac{\partial f_1}{\partial b} & \frac{\partial f_1}{\partial X_m} \\ \frac{\partial f_2}{\partial a} & \frac{\partial f_2}{\partial b} & \frac{\partial f_2}{\partial X_m} \end{bmatrix}$$

The partial derivatives can be calculated analytically as follows

$$\frac{\partial f_1}{\partial a} = \frac{\partial Y_1}{\partial a} + \frac{\partial Y_{3r}}{\partial a}$$

$$\frac{\partial f_1}{\partial b} = \frac{\partial Y_{3r}}{\partial b}$$

$$\frac{\partial f_1}{\partial X_m} = \frac{\partial Y_{2r}}{\partial X_m} = 0$$

$$\frac{\partial f_2}{\partial a} = \frac{\partial Y_{1i}}{\partial a} + \frac{\partial Y_{3i}}{\partial a}$$

$$\frac{\partial f_2}{\partial b} = \frac{\partial Y_i}{\partial b}$$

$$\frac{\partial f_2}{\partial X_m} = \frac{\partial Y_{2i}}{\partial X_m} = \frac{1}{X_m^2}$$

The expressions for $\frac{\partial Y_{1r}}{\partial a}$, $\frac{\partial Y_{3r}}{\partial a}$, $\frac{\partial Y_{3r}}{\partial b}$, $\frac{\partial Y_{1i}}{\partial a}$, $\frac{\partial Y_{3i}}{\partial a}$ and $\frac{\partial Y_{3i}}{\partial b}$ can be derived by differentiating the equation for Y_{1r} , Y_{3r} , Y_{1i} and Y_{3i} with respect to a and b.

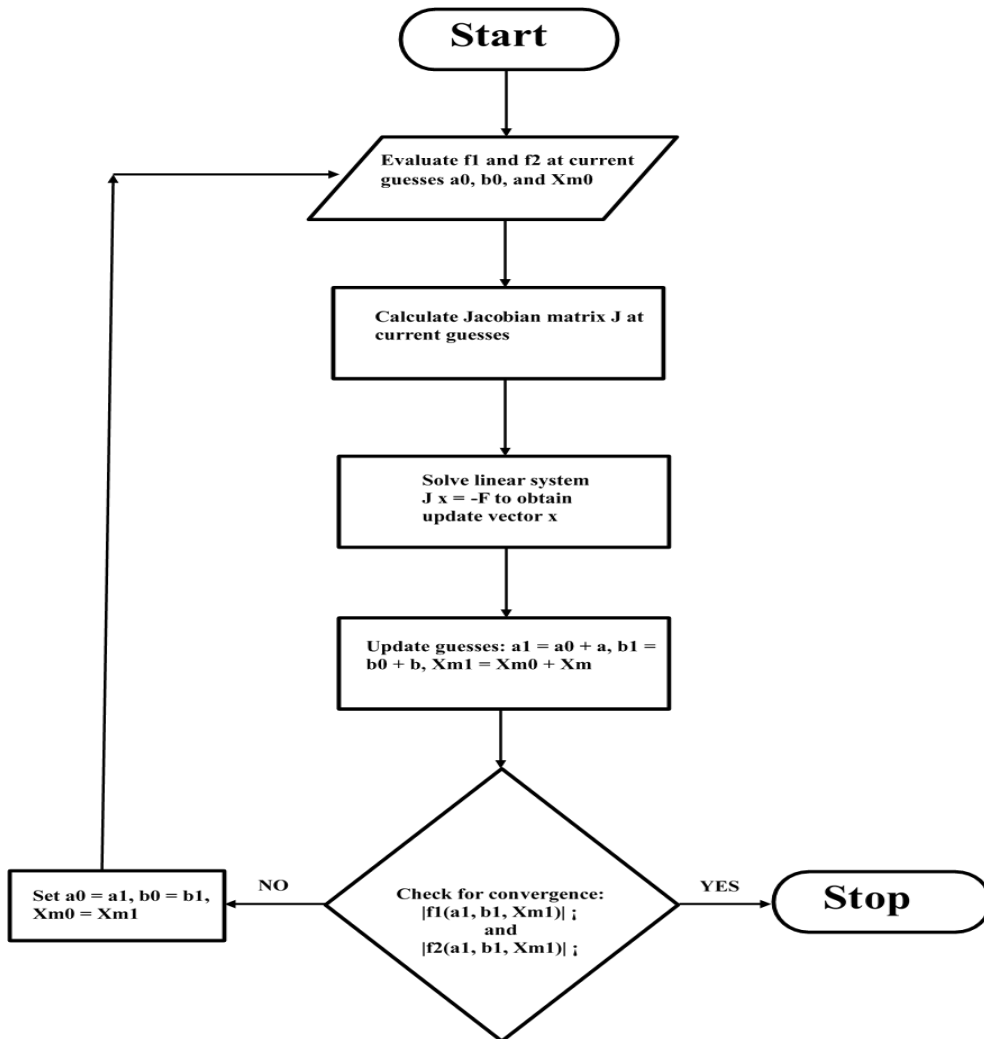


Fig. 2.6: Flow chart for start the iteration process

The detailed Newton-Raphson algorithm in mathematical form.

Initialization:

Choose initial guesses a_0, b_0, X_{m0} .

Set tolerance ε .

Set maximum iterations `max_iter`.

Set iteration counter $k = 0$.

While $k < \text{max_iter}$:

Evaluate f_1 and f_2 at a_0, b_0, X_{m0} :

$$f_1(a_0, b_0, X_{m0}) = Y_{1r}(a_0, b_0, X_{m0}) + Y_{2r}(a_0, b_0, X_{m0}) + Y_{3r}(a_0, b_0, X_{m0})$$

$$f_2(a_0, b_0, X_{m0}) = Y_{1i}(a_0, b_0, X_{m0}) + Y_{2i}(a_0, b_0, X_{m0}) + Y_{3i}(a_0, b_0, X_{m0})$$

Calculate the Jacobian matrix J at a_0, b_0, X_{m0} :

$$J = \begin{bmatrix} \frac{\partial f_1}{\partial a}(a_0, b_0, X_{m0}) & \frac{\partial f_1}{\partial b}(a_0, b_0, X_{m0}) & \frac{\partial f_1}{\partial X_m}(a_0, b_0, X_{m0}) \\ \frac{\partial f_2}{\partial a}(a_0, b_0, X_{m0}) & \frac{\partial f_2}{\partial b}(a_0, b_0, X_{m0}) & \frac{\partial f_2}{\partial X_m}(a_0, b_0, X_{m0}) \end{bmatrix}$$

Solve $J \times \Delta x = -F$ to obtain $\Delta a, \Delta b, \Delta X_m$:

$$\begin{bmatrix} \Delta a \\ \Delta b \\ \Delta X_m \end{bmatrix} = -J^{-1} \times \begin{bmatrix} f_1(a_0, b_0, X_{m0}) \\ f_2(a_0, b_0, X_{m0}) \end{bmatrix}$$

Update $a_1 = a_0 + \Delta a, b_1 = b_0 + \Delta b, X_{m1} = X_{m0} + \Delta X_m$.

If $|f_1(a_1, b_1, X_{m1})| < \varepsilon$ and $|f_2(a_1, b_1, X_{m1})| < \varepsilon$:

Stop iteration and return a_1, b_1, X_{m1} as the solution.

Else:

Set $a_0 = a_1, b_0 = b_1, X_{m0} = X_{m1}$.

Increment k by 1.

If $k \geq \text{max_iter}$:

Print "Maximum iterations reached".

CHAPTER 3

Hybrid Topology of Capacitors Excitation for Self-Excited Induction Generator

3.1 Introduction

In the previous chapter, the steady-state modelling of self-excited induction generators (SEIGs) was presented, laying the foundation for understanding their behaviour and performance characteristics. Building upon that knowledge, this chapter focuses on the development of a hybrid excitation topology that addresses the specific requirements of off-grid applications. The proposed hybrid excitation topology consists of a combination of star-connected and delta-connected excitation capacitors. The star-connected capacitors provide a common neutral point, which is essential for four-wire systems, while the delta-connected capacitors offer additional benefits, such as improved voltage regulation and reduced harmonic distortion. By combining these two configurations, the hybrid excitation topology aims to enhance the overall performance of SEIGs in off-grid applications.

To evaluate the effectiveness of the proposed topology, a comparative analysis is conducted with the conventional star-connected excitation topology. The performance of both topologies is assessed through experimental observations on a 2.2 kW, three-phase, d-series ABB make, 50 Hz, 4.8 A SEIG. The experimental setup allows for a comprehensive investigation of the behaviour and characteristics of the SEIG under various operating conditions. Furthermore, the symmetrical component technique is employed to analyse the performance of the hybrid excitation topology. This technique provides valuable insights into the behaviour of the SEIG under unbalanced loading conditions, which are common in off-grid systems. By decomposing the unbalanced voltages and currents into their symmetrical components, the analysis helps in understanding the impact of the hybrid excitation topology on the SEIG's performance.

The findings of this chapter contribute to the advancement of SEIG technology for off-grid applications. The proposed hybrid excitation topology offers a promising solution for enabling the operation of SEIGs in three-wire to four-wire systems while improving their performance and reliability. The experimental results and analysis presented in this chapter provide valuable insights for researchers and practitioners working in the field of renewable energy and off-grid power generation.

3.2 Excitation capacitor topologies

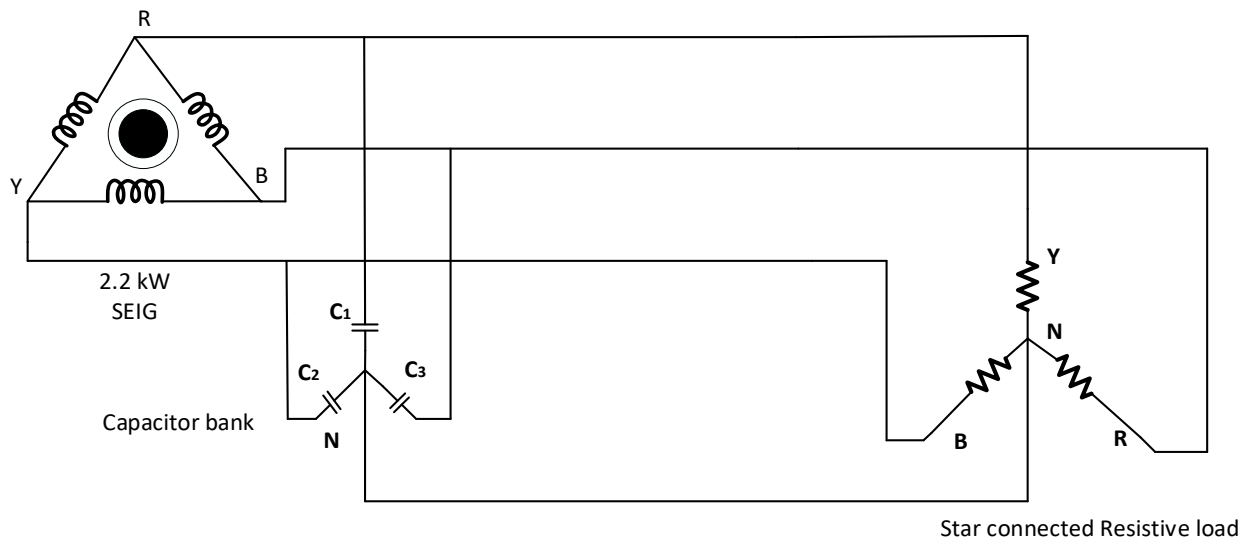


Fig. 3.1: DSS combination (delta-star-star combination)

The conventional excitation capacitor topology, known as DSS (Delta-machine, Star excitation capacitor arrangement and load), is a configuration that does not require the use of a transformer. In this topology, three capacitors are connected in a star fashion, with the star point serving as the neutral. This arrangement allows for a three-wire to four-wire system, which is suitable for off-grid applications. The DSS topology has been tested on a 2.2 kW SEIG using a 45 μF -excitation capacitor per phase, with the capacitor value determined through experimental calculations. However, the DSS topology faces challenges when subjected to extreme unbalanced loads. While balanced load conditions do not pose significant problems, as there is no substantial current flowing in the neutral wire (fourth wire), issues arise when the loads become unbalanced. This is particularly problematic for off-grid systems in rural areas, where loads are often unbalanced at all times. The unbalanced loads can lead to the flow of zero-sequence currents in the SEIG, which can negatively impact voltage regulation and overall performance. The limitations of the DSS topology under unbalanced conditions motivate the development of alternative excitation topologies that can better handle the challenges posed by off-grid applications. The hybrid excitation topology, which combines star-connected and delta-connected excitation capacitors, is one such alternative that aims to address these issues and improve the performance of SEIGs in off-grid systems.

The disadvantages of the DSS topology are

1. Poor voltage regulation under unbalanced loads: The DSS topology struggles to maintain stable voltage levels when subjected to unbalanced loads. This can lead to

voltage fluctuations and poor power quality, which can be detrimental to the connected loads.

2. Increased stress on the SEIG: The flow of zero-sequence currents in the SEIG under unbalanced conditions can cause additional stress on the machine. This stress can lead to increased heating, reduced efficiency, and potential damage to the generator over time.
3. Limited flexibility for off-grid applications: The DSS topology's susceptibility to unbalanced loads makes it less suitable for off-grid applications in rural areas, where load imbalances are common. This limitation hinders the widespread adoption of SEIGs in these settings, as the system may not perform reliably under real-world conditions.

3.3 Hybrid Combination (delta-delta/star-star combination)

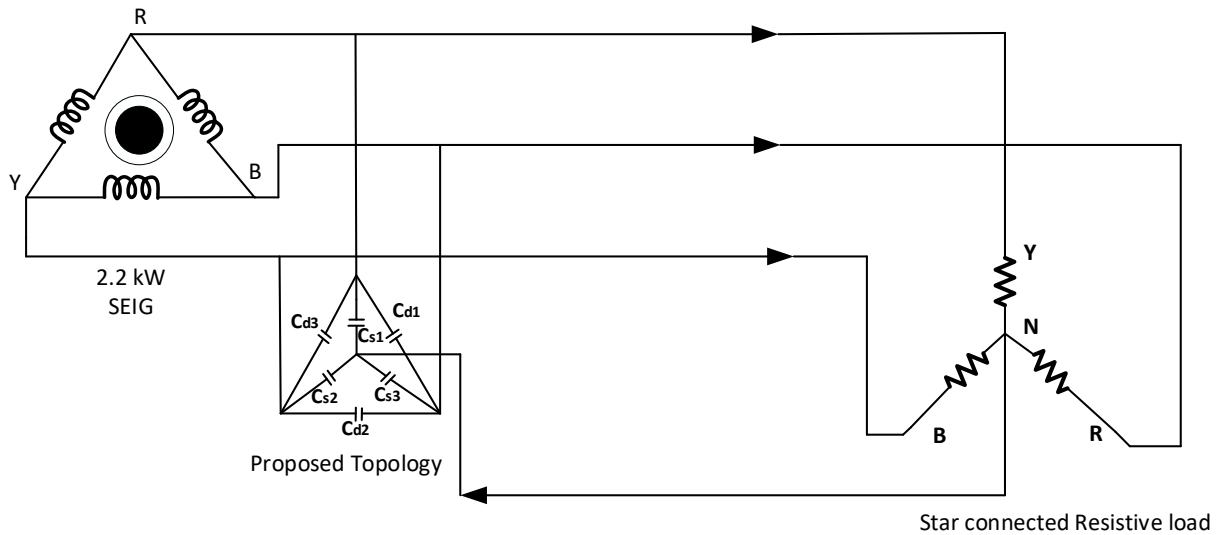


Fig. 3.2: Hybrid Combination (delta-delta/star-star combination)

The hybrid excitation topology is an innovative approach designed to address the limitations of the conventional DSS topology in off-grid applications with unbalanced loads. The key concept behind this topology is to divert the neutral current, which arises during unbalanced conditions, into all three lines of the delta-connected configuration. By doing so, the hybrid excitation topology helps in nullifying the effect of zero-sequence currents that are present during unbalanced conditions. The hybrid excitation topology combines both delta-connected and star-connected capacitors in a specific arrangement. The delta-connected capacitors are placed on one side of the SEIG, while the star-connected capacitors are placed on the other side. The neutral point of the star-connected capacitors is connected to the neutral of the generator. This unique configuration allows the neutral current to be distributed among all three

phases of the delta-connected side, effectively mitigating the impact of unbalanced loads on the SEIG. To determine the optimal values of the delta-side and star-side capacitors per phase, experimental observations were conducted. The experimental results indicate that a delta-side capacitor value of $12.5\ \mu\text{F}$ per phase and a star-side capacitor value of $2\ \mu\text{F}$ per phase provide the best performance in terms of voltage regulation and overall system stability. By effectively managing the neutral current and nullifying the effect of zero-sequence currents, the hybrid excitation topology offers several advantages over the conventional DSS topology. It improves voltage regulation under unbalanced load conditions, reduces stress on the SEIG, and enhances the system's flexibility for off-grid applications in rural areas where load imbalances are prevalent. The experimental determination of the optimal capacitor values ensures that the hybrid excitation topology is well-suited for practical implementation in off-grid systems. The combination of delta-connected and star-connected capacitors, along with the appropriate capacitor values, enables the SEIG to operate more efficiently and reliably under unbalanced load conditions, making it a promising solution for off-grid power generation in rural areas.

The hybrid excitation topology offers several advantages over the conventional DSS topology for self-excited induction generators (SEIGs) in off-grid applications. These advantages include:

1. Improved voltage regulation under unbalanced loads: By diverting the neutral current into all three lines of the delta-connected configuration, the hybrid excitation topology helps nullify the effect of zero-sequence currents that arise during unbalanced conditions. This results in better voltage regulation and stability, even when the loads are unevenly distributed across the phases.
2. Reduced stress on the SEIG: The hybrid excitation topology effectively manages the neutral current and minimizes the flow of zero-sequence currents in the SEIG. This reduction in unwanted currents leads to lower stress on the generator, including decreased heating and improved efficiency. As a result, the SEIG can operate more reliably and with a longer lifespan.
3. Enhanced flexibility for off-grid applications: The hybrid excitation topology's ability to handle unbalanced loads makes it well-suited for off-grid applications in rural areas, where load imbalances are common. This flexibility enables the widespread adoption of SEIGs in these settings, as the system can perform reliably under real-world conditions, providing a stable and efficient power supply to the connected loads.

4. Optimal capacitor values for improved performance: The experimental determination of the delta-side and star-side capacitor values per phase ensures that the hybrid excitation topology operates at its best. The optimal capacitor values of 12.5 μF for the delta side and 2 μF for the star side, as mentioned in the text, contribute to enhanced voltage regulation, system stability, and overall performance.
5. Simplified design and cost-effectiveness: The hybrid excitation topology eliminate the need for a transformer, which simplifies the overall system design and reduces the associated costs. This makes the hybrid excitation topology a cost-effective solution for off-grid power generation, especially in rural areas where infrastructure and resources may be limited.
6. Potential for scalability: The hybrid excitation topology's effectiveness in managing unbalanced loads and improving SEIG performance can be extended to larger-scale off-grid systems. As the demand for renewable energy and off-grid power generation grows, the hybrid excitation topology can be scaled up to meet the increasing power requirements while maintaining reliable operation.

3.4 Excitation calculation for SEIG

The minimum capacitance required for self-excitation in a self-excited induction generator (SEIG) can be calculated using various methods.

3.4.1 Calculation of C_{min} using name plate details

$$C_{min} = \frac{Q_g}{3 \times V^2 \times 2 \times \pi \times f}$$

$$\Delta C_{min} - \Delta SEIG$$

$$Y C_{min} - \Delta SEIG = 3 \times (\Delta C_{min} - \Delta SEIG)$$

$$\Delta C_{min} - Y SEIG$$

$$Y C_{min} - Y SEIG = 3 \times (\Delta C_{min} - Y SEIG)$$

3.4.2 Calculation of C_{min} using steady state analysis

$$C_{min} = \frac{1}{[\omega n^2 (X_{ls} + X_{smax})]}$$

$$Y C_{min} - Y SEIG$$

$$\Delta C_{min} - Y SEIG = \frac{Y C_{min} - Y SEIG}{3}$$

$$Y C_{min} - \Delta SEIG$$

$$\Delta C_{min} - \Delta SEIG = \frac{Y C_{min} - \Delta SEIG}{3}$$

3.4.3 Calculation of C min using capacitor conversion

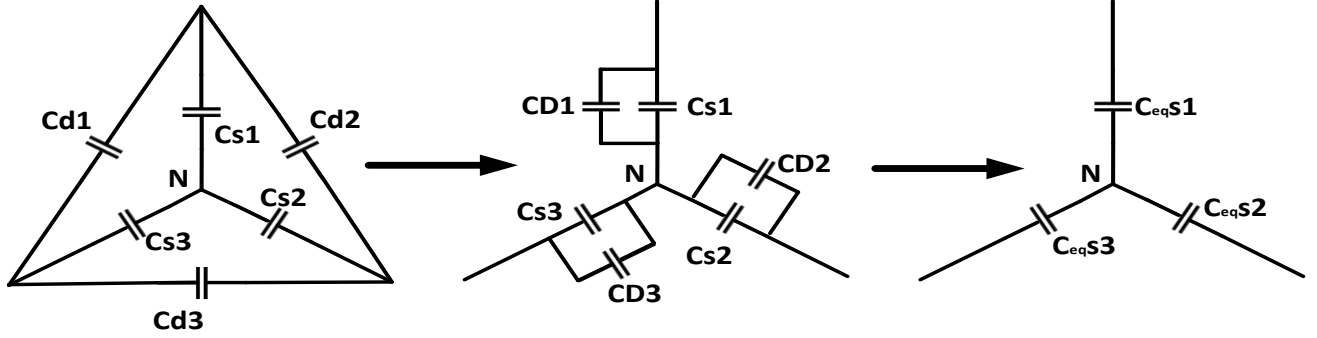


Fig. 3.3: Equivalent values of star connected capacitors

To convert the proposed hybrid configuration to an equivalent star configuration, we can use the observed values of the capacitors in the hybrid configuration. The conversion process involves calculating the equivalent capacitance values in the star configuration that would produce the same electrical characteristics as the hybrid configuration.

$$C_{D1} = \frac{C_{d1} \times C_{d2} + C_{d2} \times C_{d3} + C_{d3} \times C_{d1}}{C_{d2}}$$

$$C_{D2} = \frac{C_{d1} \times C_{d2} + C_{d2} \times C_{d3} + C_{d3} \times C_{d1}}{C_{d3}}$$

$$C_{D3} = \frac{C_{d1} \times C_{d2} + C_{d2} \times C_{d3} + C_{d3} \times C_{d1}}{C_{d1}}$$

Then the equivalent capacitor after conversion are

$$C_{eqS1} = C_{D1} + C_{S1}$$

$$C_{eqS2} = C_{D2} + C_{S2}$$

$$C_{eqS3} = C_{D3} + C_{S3}$$

The delta and star capacitors contribute to the overall reactive power in the hybrid excitation system. The total reactive power supplied from the capacitors can be calculated by considering the contributions from the delta and star configurations. Reactive Power by Delta Capacitors (Q_{Δ})

$$Q_{\Delta} = 3 \times \omega \times V_{line}^2 \times C_{\Delta}$$

Reactive Power by Star Capacitors (Q_Y)

$$Q_Y = 3 \times \omega \times V_{Phase}^2 \times C_Y$$

Where

ω is the angular frequency.

V_{line} is the line voltage for delta.

V_{Phase} is the phase voltage for star.

$$Q_{total} = Q_{\Delta} + Q_Y$$

This total reactive power is less than the reactive power in the case of conventional star configuration. Hence, more active power can be transfer to loads.

3.5 Methods for evaluation

To test the effectiveness of proposed configuration over the existing one, the system is examined using the symmetrical components. Symmetrical components are not a new technique, but one the bests for analysis of unbalance condition.

$$I_{positive} = \frac{1}{3} (I_A + aI_B + a^2I_C)$$

$$I_{negative} = \frac{1}{3} (I_A + a^2I_B + aI_C)$$

$$I_{zero} = \frac{1}{3} (I_A + I_B + I_C)$$

where $a = e^{j120^\circ}$ is the symmetrical operator representing, and I_A , I_B & I_C are line current at the load side.

During balance condition, when all three phases are drawing equal amount of currents, then with the use of symmetrical components it is known that the total current flowing through neutral wire is zero.

$$I_A + I_B + I_C = 0$$

During unbalance operations, which is obvious in the rural areas, the total sum of current following through neutral wire will not be zero.

$$I_A + I_B + I_C \neq 0$$

The negative and zero-sequence currents are components of the unbalanced currents. Another way to test the degree of unbalance is by using graph of the ratio of negative to positive sequence or ratio of zero sequence components to positive sequence component. The ratio must be close to unity as positive sequence components represents the perfect balance condition.

$$\frac{I_{negative}}{I_{positive}} \text{ or } \frac{I_{zero \text{ Sequence}}}{I_{positive}} = 1$$

Hardware testing is done with three phase Squirrel Case Induction Machine with 2.2 kW, 50 Hz, 4.8 A, D-series ABB make coupled with three phase 3.3 kW Squirrel Case Induction Machine, 50 Hz, 11.1 A, ABB make driven by 3.3 KW Three phase Variable Frequency Drive (VFD). Incandescent lamps of various rating are used for three phase loads.

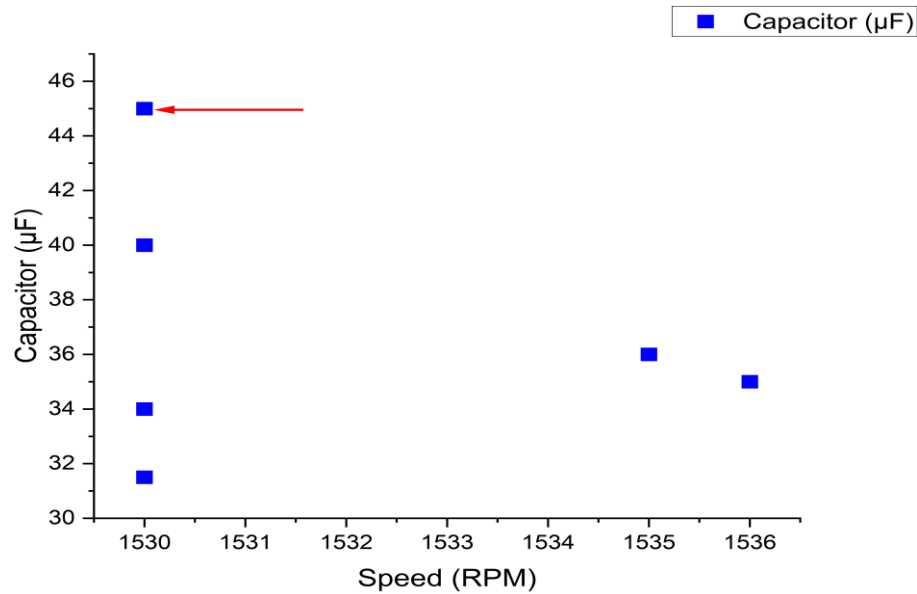


Fig. 3.4: Capacitor selection on DSS condition

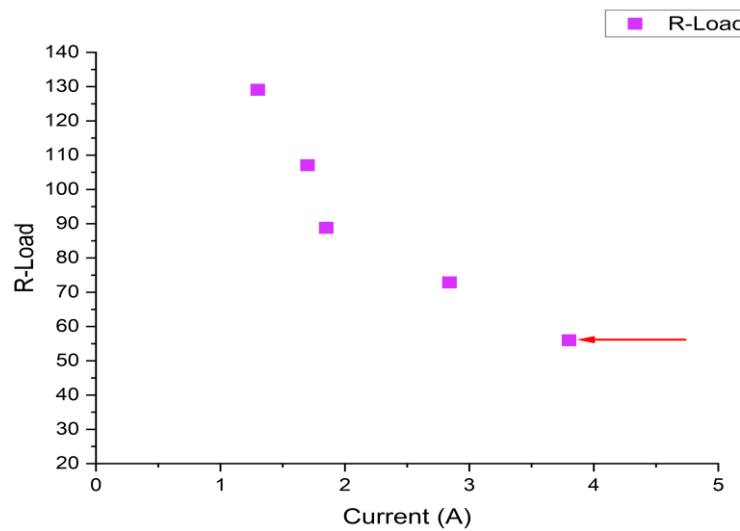


Fig. 3.5: current and load on DSS condition

Experiments have shown that the values of excitation capacitors for 50 Hz with standard terminal voltage is 45 μF . This capacitance value provides a higher loading capacity and helps the SEIG system operate more efficiently. And second the machine is run with the proposed configuration, for no-load condition for finding the best possible combination of star and delta capacitor value.

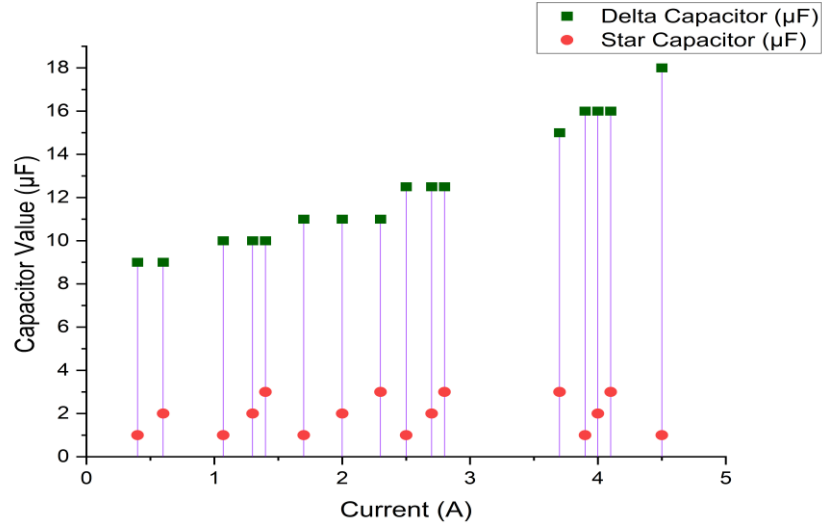


Fig. 3.6: Hybrid capacitor combination in Delta and Star

The corresponding value of star and delta capacitors for 50 Hz, at -2% slip and terminal voltage 415 V is 2.0 μF and 12.5 μF respectively. Selecting the right capacitance values is crucial in maintaining the stable and reliable operation of the generator. The capacitance value of 12.5 μF , in the delta configuration along with 2 μF in the star configuration contribute to a efficient excitation system for the SEIG. This ensures that the SEIG operates safely and within desired limits. Further, adjusting these capacitance values potential issues, like frequency deviations and voltage instability can be minimized.

CHAPTER 4

Experimental setup discretion

CHAPTER 4

Experimental setup discretion

4.1 Introduction

In the previous chapter, a hybrid excitation topology combining star and delta connections was introduced for self-excited induction generators (SEIGs) in off-grid applications. The proposed topology aimed to facilitate the operation of SEIGs in three-wire to four-wire systems while enhancing their performance and reliability. Building upon the theoretical foundation established in Chapter 3, this chapter focuses on the experimental setup and investigation of various configurations of SEIGs, excitation capacitors, and loads.

The experimental study encompasses four distinct combinations: star-star-star, delta-delta-star, delta-star-star, and star-delta-star. These configurations represent different arrangements of the SEIG, excitation capacitors, and load connections. The objective is to determine the optimal configuration that yields the best performance characteristics for the SEIG. In the star-star-star combination, the SEIG, excitation capacitors (40 and 45 microfarads), and load are all connected in a star configuration. The experimental results demonstrate that at frequencies of 50 and 52 hertz, the SEIG generates output voltages of 160 and 180 volts, respectively, with corresponding load values of 70 and 60 ohms. The second combination, delta-delta-star, consists of a delta-connected SEIG, delta-connected excitation capacitors (12 and 15 microfarads), and a star-connected load. At frequencies of 49 and 47 hertz, the SEIG produces output voltages of 175 and 145 volts, respectively, with load values of 80 and 60 ohms. The third combination, delta-star-star, employs a delta-connected SEIG, hybrid-connected excitation capacitors (delta 40 microfarads, star 2 microfarads) and (delta 45 microfarads, star 2 microfarads), and a star-connected load. Similar to the second combination, at frequencies of 49 and 47 hertz, the SEIG generates output voltages of 175 and 145 volts, respectively, with load values of 80 and 60 ohms. In the fourth combination, star-delta-star, the SEIG is connected in a star configuration, while the excitation capacitors (12 and 15 microfarads) are connected in delta, and the load is connected in star. The experimental results show that at frequencies of 49 and 47 hertz, the SEIG produces output voltages of 180 and 150 volts, respectively, with load values of 80 and 60 ohms. To determine the capacitor values for the delta-star-star combination, a specific experimental setup is employed. As illustrated in the figure 4.1, the setup consists of a variable frequency drive (VFD) (1), excitation capacitors (2), a Fluke power analyser (3), a neutral current clamp (4), a 2.2 kW SEIG (5), a prime mover (6) for rotating the

SEIG, an R-load (7), and a stator current clamp (8). The experiment is conducted at a slip of -2 percent, corresponding to a rotational speed of 1500 rpm. The experimental investigation presented in this chapter provides valuable insights into the performance characteristics of SEIGs under different configurations. By analysing the voltage and load values at various frequencies, the optimal combination can be identified. The experimental setup description and the detailed results contribute to a comprehensive understanding of the practical implementation and behaviour of SEIGs in off-grid applications.

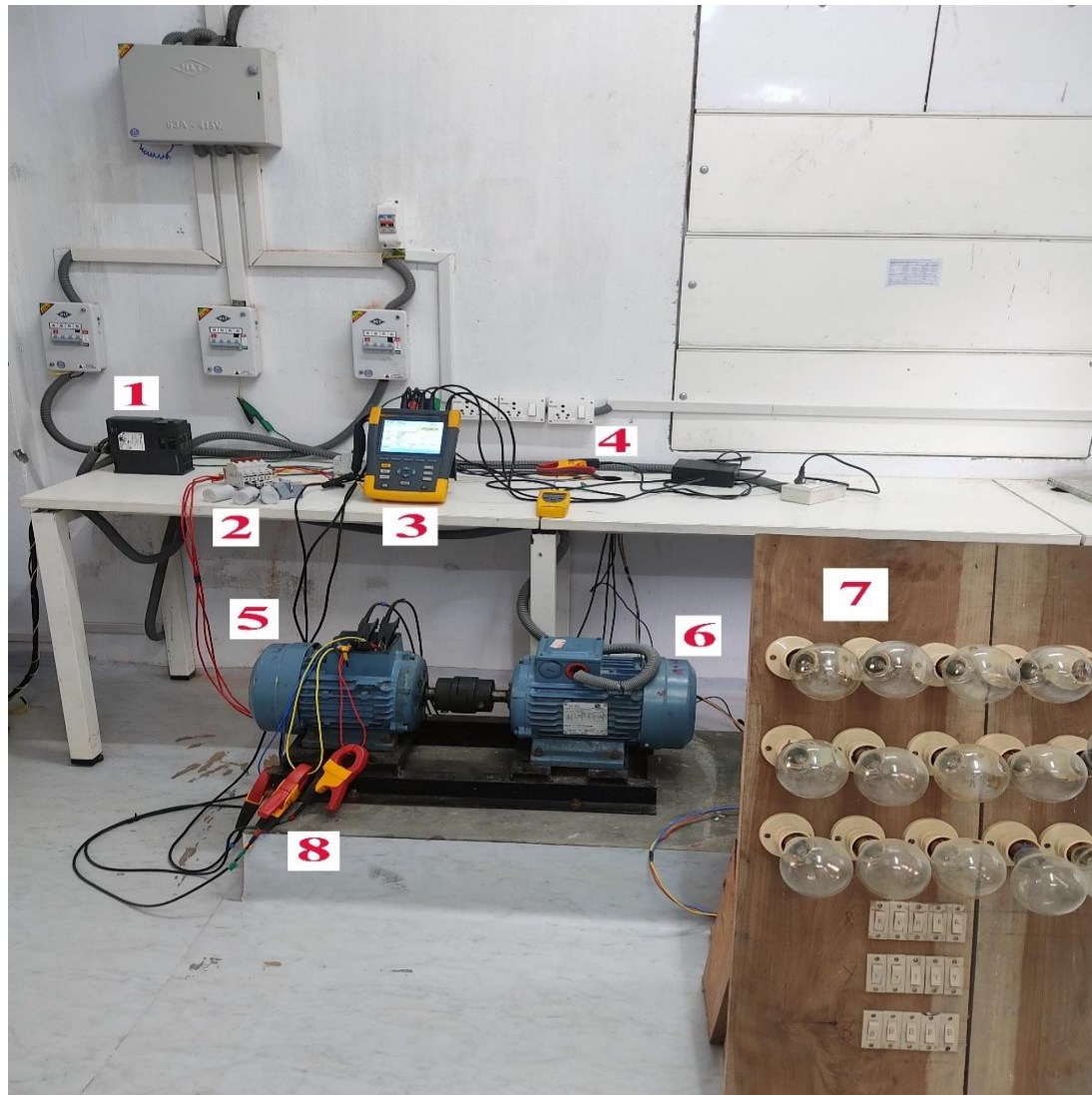


Fig. 4.1: Hardware set-up of Δ -Y-Y combination

Table 1 presents a comparative analysis of different machine, capacitor, and load configurations for a self-excited induction generator (SEIG) setup. The table includes four distinct combinations: star-star-star, delta-delta-star, delta-star-star, and star-delta-star. For each combination, the capacitor values in microfarads (μF) are provided for both delta (Δ) and star (Y) connections, along with the corresponding operating frequency (Hz),

output voltage (V), and connected resistive load (R-Load) in ohms (Ω). The star-star-star combination employs capacitor values of 40 μF and 45 μF connected in a star, resulting in output voltages of 160 V and 180 V at frequencies of 50 Hz and 52 Hz, respectively, when connected to loads of 70 Ω and 60 Ω . In the delta-delta-star combination, capacitor values of 12 μF and 15 μF are connected in delta, yielding output voltages of 175 V and 145 V at frequencies of 49 Hz and 47 Hz, respectively, with loads of 80 Ω and 60 Ω . The delta-star-star combination utilizes a hybrid capacitor configuration, with 40 μF delta and 2 μF star capacitors generating 196 V at 50 Hz with a load of 72.9 Ω , while 45 μF delta and 2 μF star capacitors produce 228 V at 50 Hz with a load of 56 Ω . Lastly, the star-delta-star combination employs capacitor values of 12 μF and 15 μF connected in delta, resulting in output voltages of 180 V and 150 V at frequencies of 49 Hz and 47 Hz, respectively, when connected to loads of 80 Ω and 60 Ω .

Table I Combination of machine, excitation and load

S.No	Combination name (Machine – capacitor-Load)	Capacitor		Frequency (Hz)	Voltage (V)	R- Load (Ω)
		Δ (μf)	Y (μf)			
1 .	Star-star- star	--	40	50	160	70
		--	45	52	180	60
2.	Delta-Delta – Star	12	--	49	175	80
		15	--	47	145	60
3.	Delta- Star- Star	40	2	50	196	72.9
		45	2	50	228	56
4.	Star –Delta –Star	12	--	49	180	80
		15	--	47	150	60

Table 2 presents the experimental results for the delta-star-star combination of a self-excited induction generator (SEIG) at different speeds (RPM). This configuration was identified as the best combination based on the findings from Table 1. The experiment aims to investigate the performance of the SEIG under varying speed conditions while maintaining the optimal machine, capacitor, and load configuration. The table provides information on the rotational speed, capacitance values for delta (Δ) and star (Y) connected capacitors, voltage drop, output current, operating frequency, and connected resistive load for each speed.

Table II Testing of best combination with different load

Speed (RPM)	Capacitor (μF) Δ/Y	Volt Drop(V)	Current (A)	Frequency (Hz)	R-Load(Ohms)
1536	35/2	163	1.85	49.9	88.8
1530	34/2	170	1.7	50	107.1
1530	31.5/2	169.78	1.3	50	129.1
1535	36/2	179.16	2.03	50	89.8
1530	40/2	196	2.84	50	72.9
1530	45/2	228	3.8	50	56

Analysing the data, it is observed that as the speed increases from 1530 RPM to 1536 RPM, the voltage drop increases from 169.78 V to 179.16 V, while the current decreases from 1.3 A to 2.03 A, and the frequency remains constant at 50 Hz. Further increasing the speed to 1536 RPM results in a voltage drop of 163 V, a current of 1.85 A, and a frequency of 49.9 Hz, with a connected load of 88.8 Ω . At the highest speed of 1530 RPM, with capacitor values of 45 μF (Δ) and 2 μF (Y), the SEIG achieves a voltage drop of 228 V, a current of 3.8 A, and a frequency of 50 Hz, with a connected load of 56 Ω . These experimental results demonstrate the impact of speed variation on the performance characteristics of the SEIG in the delta-star-star configuration. As the speed increases, the voltage drops and current exhibit corresponding changes, while the frequency remains relatively stable around 50 Hz. The findings suggest that the delta-star-star combination, with appropriate capacitor values, can provide satisfactory performance across a range of speeds, maintaining a consistent frequency close to the desired 50 Hz, indicating its suitability for off-grid applications.

4.2 Calculation of parameters

The following testing and parameter calculation is done

1. Continuity test
2. Short circuited test between the windings
3. Earth test or short-circuited between windings and body of motor
4. Block rotor test
5. No load tests

6. Magnetic saturation experiment

4.2.1 Continuity test

The terminal box of the SEIG is as shown in fig. 4.2 The SEIG has three balanced stator windings, each consisting of two terminals: W1 & W2, U1 & U2, and V1 & V2. These windings are connected to the terminal box of the motor. To perform the continuity test, a multi-meter is used. The test is done by connecting the two probes of the multi-meter to the terminals of each winding. When the probes are connected to the terminals of the same winding (e.g., W1 & W2, U1 & U2, or V1 & V2), the multi-meter should produce a beeping sound, indicating that there is continuity (i.e., a closed circuit) within that winding.

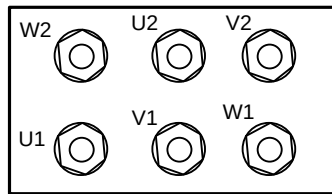


Fig. 4.2: Terminal box of the SEIG

However, when the probes are connected to terminals of different windings (e.g., W1 & U1, W1 & U2, W1 & V1, W1 & V2, etc.), the multi-meter should not produce a beeping sound. If a beep is heard when connecting terminals of different windings, it would indicate a short circuit between those windings, which is a fault condition.

4.2.2 Short-circuited and earth test

Short-circuited test

The short-circuited test is conducted to check if there are any short circuits between the different stator windings of the induction motor. A short circuit between windings can cause severe damage to the motor and pose a safety risk. To perform the short-circuited test, a multi-meter is used to test the continuity between the different windings. The test is done by connecting one probe of the multi-meter to a terminal of one winding (e.g., W1) and the other probe to a terminal of another winding (e.g., U1, U2, V1, or V2). This process is repeated for all possible combinations of windings. If the multi-meter produces a beeping sound or shows continuity between any two windings, it indicates a short circuit fault, and the motor should not be operated until the issue is resolved.

Earth test

The earth test, also known as the insulation resistance test, is performed to ensure that the stator windings are properly insulated from the motor's body or frame. Proper insulation is essential to prevent electric shock hazards and protect the motor from damage. To conduct the earth test, a multi-meter is used to check the continuity between each stator winding and the motor's body or frame. The test is performed by connecting one probe of the multi-meter to a terminal of a winding (e.g., W1, U1, or V1) and the other probe to the motor's body or frame. If the multi-meter shows continuity or produces a beeping sound during the earth test, it indicates that the insulation between the winding and the motor's body is compromised, and the motor should not be operated until the insulation issue is resolved.

4.2.3 Blocked rotor test

The blocked rotor test is a standard test performed on induction machines, including self-excited induction generators (SEIGs), to determine their equivalent circuit parameters, specifically the rotor resistance and leakage reactance. This test is crucial for understanding the machine's performance and for accurate modelling and simulation.

1. Test Setup:

- The rotor of the SEIG is mechanically locked or blocked to prevent rotation.
- A variable voltage source, such as an autotransformer or a variac, is connected to the stator windings of the SEIG.
- Appropriate measuring instruments, such as voltmeters, ammeters, and wattmeter, are connected to measure the voltage, current, and power during the test.

2. Test Procedure:

- The variable voltage source is adjusted to apply a reduced voltage to the stator windings, typically around 25% to 50% of the rated voltage. This is done to limit the current drawn by the machine during the test.
- The voltage is slowly increased until the rated current is reached in the stator windings.
- At this point, the voltage, current, and power readings are recorded.
- The test is repeated for all three phases of the SEIG to ensure consistency.

3. Equivalent Circuit Parameters:

- The blocked rotor test helps determine the following equivalent circuit parameters:
- Rotor Resistance (R_2): The rotor resistance represents the electrical resistance of the rotor windings referred to the stator side.

- Leakage Reactance (X2): The leakage reactance represents the inductive reactance of the rotor windings referred to the stator side.

4. Calculations:

To determine the series parameters of the motor blocked rotor test was done, this was done with two wattmeter's method, where the rotor was kept blocked and motor was run with rated current of 4.8 A. And the measured value of power is

The values observed are

$P_{sc} = 281 \text{ Watt}$, $I_{sc} = 4.8 \text{ A}$, $V_{sc} = 125 \text{ Volt}$.

Hence, $R_{eq} = \frac{281}{3 \cdot 4.8^2} = 4.0653 \Omega$

$$Z_{eq} = \frac{125/\sqrt{3}}{4.8} = 15.0352 \Omega$$

Therefore, $X_{eq} = \sqrt{15.0352^2 - 4.0653^2} = 14.4751 \Omega$

$$\text{And } X_s = X_r = 14.4751/2 = 7.2376 \Omega$$

$$L_{lr} = L_{ls} = 7.2376 / (2 \cdot \pi \cdot 50) = \mathbf{0.0230 \text{ H}}$$

5. Parameter Referral:

- The calculated resistance and reactance values are referred to the stator side using the appropriate turns ratio between the stator and rotor windings.
- The referred rotor resistance (R2) and leakage reactance (X2) are obtained.

6. Interpretation and Use:

- The determined rotor resistance (R2) and leakage reactance (X2) are used in the equivalent circuit model of the SEIG.
- These parameters are essential for predicting the performance characteristics of the SEIG, such as voltage regulation, efficiency, and torque-speed behaviour.
- The blocked rotor test results are also used for designing appropriate control strategies and protection systems for the SEIG.

4.2.4 No load test

The no-load test is another important test conducted on self-excited induction generators (SEIGs) to determine their equivalent circuit parameters, specifically the magnetizing reactance and core loss resistance. This test provides insights into the SEIG's behaviour when operating at no load or under light load conditions.

1. Test Setup:

- The SEIG is mechanically coupled to a prime mover, such as a diesel engine or a DC motor, which drives the generator at its rated speed.
- The stator windings of the SEIG are connected to a capacitor bank of appropriate value to provide the necessary reactive power for self-excitation.
- Measuring instruments, such as voltmeters, ammeters, and wattmeter, are connected to measure the voltage, current, and power during the test.

2. Test Procedure:

- The prime mover is started, and the SEIG is brought up to its rated speed.
- The capacitor bank is adjusted to achieve the desired voltage level at the stator terminals, typically the rated voltage of the SEIG.
- Once the SEIG reaches a steady-state condition, the voltage, current, and power readings are recorded.
- The test is repeated for different capacitance values to obtain a range of data points.

3. Equivalent Circuit Parameters:

- The no-load test helps determine the following equivalent circuit parameters:
- Magnetizing Reactance (X_m): The magnetizing reactance represents the inductive reactance of the magnetizing branch in the equivalent circuit model.
- Core Loss Resistance (R_c): The core loss resistance represents the resistance that accounts for the power losses in the SEIG's core due to hysteresis and eddy currents.

4. Calculations:

In this test, motor is run on no load at rated voltage (415 Volt) and rated frequency (50Hz). Two wattmeter's method is used to this test.

$P_o = 118$ Watt, $I_o = 0.51$ A, $V_o = 415$ Volt.

$$\cos \theta_o = 118 / (\sqrt{3} * 415 * 0.51) = 0.3218$$

$$\sin \theta_o = 0.9468$$

$$I_m = \sin \theta_o * 0.51 = 0.4829 \text{ A}$$

$$X_m = \frac{415 / \sqrt{3}}{0.4829} = 496.16 \Omega$$

Therefore,

$$L_m = X_m / 2 * \pi * f = \mathbf{1.5793 \text{ H}}$$

5. Parameter Determination:

- The magnetizing reactance (X_m) is calculated using the formula $X_m = V / (\sqrt{3} \times I_o \times \sin \phi_0)$, where $\sin \phi_0 = \sqrt{(1 - \cos^2 \phi_0)}$.

- The core loss resistance (R_c) is calculated using the formula $R_c = V^2 / P_0$.

6. Interpretation and Use:

- The determined magnetizing reactance (X_m) and core loss resistance (R_c) are used in the equivalent circuit model of the SEIG.
- The magnetizing reactance (X_m) represents the reactive power required for self-excitation and the establishment of the magnetic field in the SEIG.
- The core loss resistance (R_c) accounts for the power losses in the SEIG's core, which affect the efficiency and performance of the generator.
- The no-load test results are used in conjunction with the blocked rotor test results to develop a complete equivalent circuit model of the SEIG.

4.2.5 Magnetic saturation experiment

1. Equipment setup:

- The 2.2 kW three-phase induction motor (acting as the SEIG) is directly coupled with a 3.3 kW three-phase induction motor, which serves as the prime mover.
- The prime mover is chosen to have a slightly higher rating than the SEIG to ensure smooth and reliable operation.

2. Individual testing:

- Before coupling the motors, each motor is tested individually to ensure they rotate in the clockwise direction.
- The 2.2 kW motor's speed is controlled using an autotransformer, while the prime mover's speed is controlled by a 3.3 kW variable frequency drive (VFD).

3. Coupling and operation:

- After confirming the identical rotations of both rotors, the SEIG is coupled with the prime mover.
- The prime mover is supplied with rated voltage and frequency to drive the SEIG at its rated speed.

4. Magnetization curve measurement:

- An autotransformer is connected to the stator of the SEIG.
- The voltage from the autotransformer is gradually increased while observing the voltage and current at the stator.
- The corresponding voltage and current values are recorded at each step.

5. Data plotting:

- The recorded voltage and current values are used to plot a graph, which represents the magnetization curve of the SEIG.
 - The graph typically shows the relationship between the stator voltage and the magnetizing current.
6. Experiment repetition:
- The same procedure is repeated for a 5.5 kW SEIG coupled with a 7.5 kW prime mover.

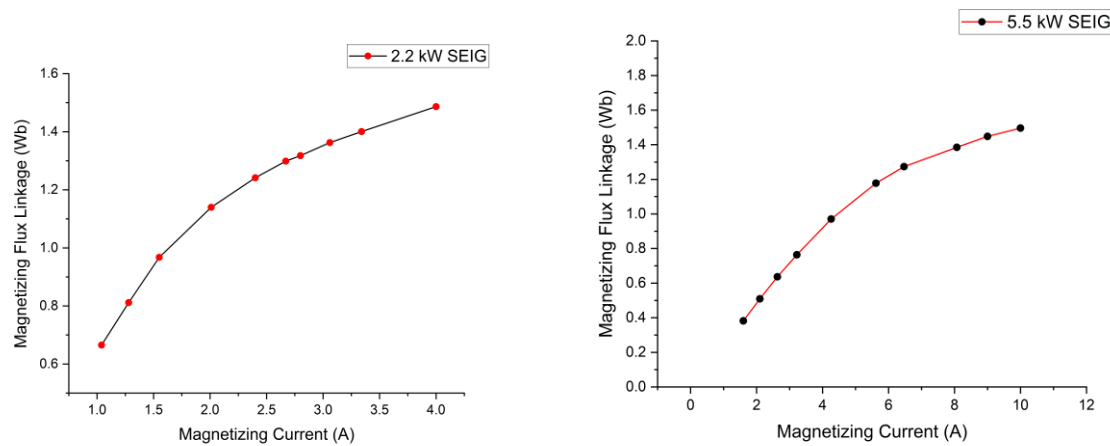


Fig. 4.3: Saturation Curves for both the machine, 2.2 & 5.5 kW SEIGs

The magnetic saturation experiment is crucial for understanding the magnetization characteristics of the SEIG. The plotted graph provides insights into the following aspects:

1. Magnetization curve: The graph shows the relationship between the stator voltage and the magnetizing current. It helps determine the saturation point of the SEIG, beyond which the magnetizing current increases rapidly with a small increase in voltage.
2. Knee point: The knee point on the magnetization curve indicates the onset of magnetic saturation. It is the point where the curve transitions from the linear region to the saturated region.
3. Saturation region: The saturation region of the magnetization curve represents the operating range where the SEIG's core is heavily saturated. In this region, a small increase in voltage leads to a significant increase in magnetizing current.
4. Air gap line: The linear portion of the magnetization curve, before reaching the knee point, is known as the air gap line. It represents the unsaturated region of operation.

The magnetization curve obtained from the magnetic saturation experiment is essential for accurate modelling and simulation of the SEIG. It helps in determining the appropriate value of the magnetizing reactance (X_m) in the equivalent circuit model.

CHAPTER 5

Hybrid Excitation based Parallel Operation of Self-Excited Induction Generators for Rural Applications

CHAPTER 5

Hybrid Excitation based Parallel Operation of Self-Excited Induction Generators for Rural Applications

5.1 Introduction

In the previous chapter, discussed the experimental setup and investigated various combinations of machine configuration, excitation capacitor connection, and load connection for a self-excited induction generator (SEIG) setup. The experimental results provided valuable insights into the performance characteristics of SEIGs under different configurations, with the delta-star-star combination exhibiting promising results. Building upon these findings, this chapter focuses on the development and analysis of a hybrid excitation system for parallel-connected SEIGs, specifically designed for rural electrification applications. Rural electrification poses unique challenges, such as the need for reliable and efficient power generation using locally available renewable energy sources. SEIGs have emerged as a viable solution due to their simple construction, low cost, and ability to generate electricity without requiring an external power grid. However, to meet the growing energy demands of rural communities, it is often necessary to operate multiple SEIGs in parallel. Parallel operation of SEIGs enables increased power generation capacity and provides redundancy in case of individual machine failure.

This chapter presents a novel hybrid excitation system for parallel-connected SEIGs with different power ratings, namely 2.2 kW and 5.5 kW. The proposed system aims to enhance the performance of the parallel-connected SEIGs by utilizing separate excitation capacitor banks for each machine. The generators are connected in a delta configuration, while the excitation capacitors are arranged in a hybrid star-delta configuration. This unique arrangement provides a neutral point and allows for the examination of load-sharing capability under various load conditions, including resistive, inductive-resistive, and nonlinear loads. To analyse the performance of the parallel-connected SEIG system with the hybrid excitation, a state-space model is developed. The Runge-Kutta method, a numerical integration technique, is employed to solve the state-space model, enabling a comprehensive analysis of key performance indicators such as voltage regulation, load sharing, and efficiency. The use of the Runge-Kutta method provides a reliable and accurate means of simulating the system's behaviour, facilitating the optimization of the hybrid excitation system. Experimental results presented in this chapter demonstrate the superiority of the hybrid excitation system over the conventional star-connected excitation commonly used in SEIG systems. The proposed system achieves significant efficiency improvements of 3.5%, 4.2%, and 2.9% under resistive, inductive-

resistive, and nonlinear load conditions, respectively. Additionally, the hybrid excitation system exhibits better voltage regulation, with a maximum deviation of only 1.8%, and maintains accurate load sharing, with an error of less than 5% across all tested scenarios. The findings of this chapter highlight the potential of the hybrid excitation system for enhancing the performance and efficiency of parallel-connected SEIGs in rural electrification applications. By effectively utilizing renewable energy sources and improving power quality, the proposed system offers a capable solution for meeting the electricity needs of remote areas. The Runge-Kutta method proves to be a valuable tool for analysing and optimizing the hybrid excitation system, ensuring its reliable and efficient operation.

5.2 System Configuration

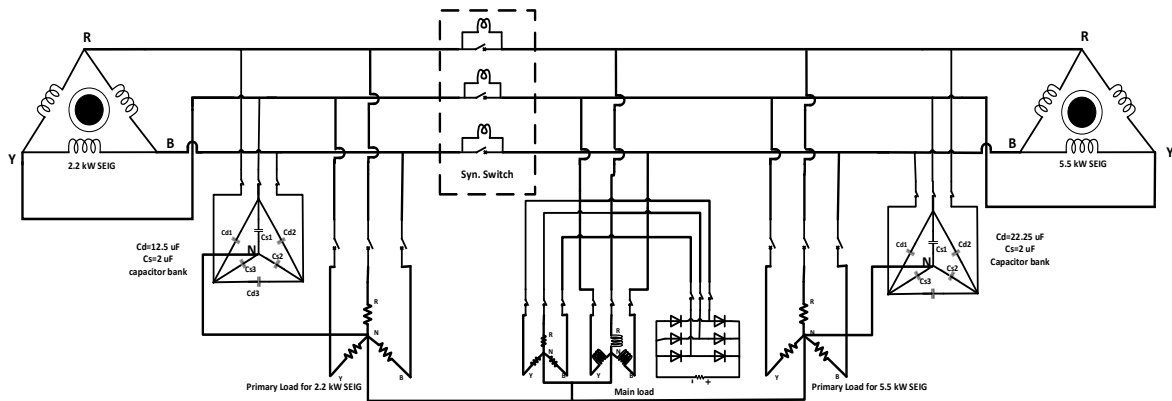


Fig. 5.1: Proposed method for parallel connect two SEIGs with Hybrid Excitation.

The schematic diagram presented illustrates a parallel operation system that combines two self-excited induction generators (SEIGs) with different power ratings: a 2.2kW SEIG on the left side and a 5.5kW SEIG on the right side. The primary objective of this arrangement is to harness the combined power generation capabilities of both SEIGs, enabling the system to meet higher load demands and enhance overall reliability.

One of the key features of this parallel operation system is the use of hybrid-connected capacitor banks for each SEIG. These capacitor banks play a crucial role in providing the necessary reactive power required for the self-excitation of the induction generators. By employing a hybrid connection, which typically involves a combination of series and parallel capacitor configurations, the system can achieve optimal excitation and improve the performance of the SEIGs. In addition to the hybrid-connected capacitor banks, each SEIG is also connected to its primary load. The primary load serves an essential purpose in stabilizing the operation of the SEIG before it is synchronized with the parallel system. By having a dedicated load connected to each generator, the SEIGs can achieve stable operating conditions, ensuring smooth

integration into the parallel arrangement. The synchronization of the two SEIGs is facilitated by a synchronous switch, which acts as the central component in the parallel operation system. The synchronous switch is responsible for precisely aligning the voltages, frequencies, and phase angles of the two generators, allowing them to operate in harmony. This synchronization process is critical to ensure seamless power sharing and avoid any potential conflicts or instabilities that may arise from mismatched generator parameters. Once the SEIGs are synchronized and connected through the synchronous switch, they can effectively contribute to the power supply of the main load. The main load in this system comprises three distinct components: a resistive load (R), an inductive load (L), and a non-linear load. The resistive load represents the purely resistive elements in the system, such as heating elements or incandescent lamps. The inductive load, on the other hand, includes components that exhibit inductive properties, such as motors or transformers. Lastly, the non-linear load encompasses devices with non-linear voltage-current characteristics, such as power electronic converters or rectifiers. The parallel operation of the 2.2kW and 5.5kW SEIGs allows for the efficient utilization of their combined power generation capacity. By distributing the load demand across both generators, the system can achieve improved power quality, enhanced reliability, and increased overall efficiency. The parallel arrangement also provides redundancy, ensuring that if one SEIG experiences a fault or requires maintenance, the other SEIG can continue to supply power to the main load, minimizing disruptions. To ensure the optimal performance and stability of the parallel operation system, various control and protection mechanisms are typically implemented. These may include voltage and frequency regulation, load-sharing control, and fault protection schemes. By properly coordinating the operation of the SEIGs and managing the power flow, the system can maintain a stable and reliable power supply to the main load.

5.3 Procedure for parallel connection of two different rated SEIGs.

1. Initial setup:
 - Connect the 2.2 kW and 5.5 kW SEIGs to the synchronizing switch and the load, as shown in Fig. 5.1.
 - Ensure that the synchronizing switch is in the OFF position to isolate the two SEIGs from each other during the initial setup.
2. Individual SEIG operation:
 - Start by running both SEIGs separately, gradually adjusting the slip of each machine.
 - Slowly change the slip of the 2.2 kW SEIG until its terminal voltage reaches slightly higher than the standard value of 415 V and its frequency stabilizes at 50 Hz.

- Similarly, adjust the slip of the 5.5 kW SEIG until its terminal voltage is slightly above 415 V and its frequency is at 50 Hz.
 - Monitor the voltage and frequency of each SEIG using appropriate measuring instruments to ensure they are within the desired range.
3. Starting load connection:
- Once both SEIGs have reached the desired voltage and frequency, switch ON the starting loads for each SEIG individually.
 - The starting loads help to stabilize the operation of the SEIGs before synchronization and parallel connection.
 - Ensure that the starting loads are properly sized and connected to their respective SEIGs.
4. Voltage and frequency matching:
- To prepare for synchronization, gradually match the voltage and frequency of the two SEIGs by adjusting the slip of one of the machines.
 - Choose one SEIG (either the 2.2 kW or 5.5 kW) and carefully adjust its slip until its voltage and frequency closely match those of the other SEIG.
 - Continuously monitor the voltage and frequency of both SEIGs during this process to ensure they remain within acceptable limits.
5. Synchronization:
- Once the voltages and frequencies of both SEIGs are closely matched, proceed with the synchronization process.
 - Verify that the phase sequence and phase angles of the two SEIGs are aligned using a synchronizing device or by observing the synchronizing lamps.
 - When the synchronizing conditions are met (equal voltages, frequencies, and phase angles), quickly close the synchronizing switch to connect the two SEIGs in parallel.
 - The synchronizing switch should be rated to handle the combined power of both SEIGs and should be capable of closing rapidly to minimize synchronization transients.
6. Main load connection:
- After successfully synchronizing and paralleling the two SEIGs, switch ON the main load that the parallel-connected SEIGs will supply power to.
 - The main load can be a combination of resistive, inductive, and non-linear components, as mentioned in the diagram explanation.

- Gradually increase the main load while monitoring the performance of the parallel-connected SEIGs.
7. Disconnecting starting loads:
- Once the main load is successfully connected and the parallel-connected SEIGs are operating stably, the starting loads can be disconnected from each SEIG.
 - Carefully switch OFF the starting loads for the 2.2 kW and 5.5 kW SEIGs, ensuring that the main load remains connected to the parallel system.
8. Monitoring and control:
- Continuously monitor the operation of the parallel-connected SEIGs, including their voltages, frequencies, power output, and load sharing.
 - Implement necessary control mechanisms, such as voltage and frequency regulation, to maintain the stable operation of the parallel system.
 - Regularly check the synchronizing switch, capacitor banks, and other components for any signs of abnormalities or faults.

5.4 Hardware set-up of Parallel combination

The hardware setup presented in the figure 5.2 is designed for the parallel operation of two Self-Excited Induction Generators (SEIGs) with different capacities: 2.2 kW and 5.5 kW. The purpose of this setup is to study the behaviour and performance of the parallel-connected SEIGs under various linear and non-linear loading conditions. The 2.2 kW SEIG subsystem consists of a prime mover (6) coupled with an induction generator (5). The voltage of the prime mover is regulated by a Variable Frequency Drive (VFD) (1). A capacitor (3) is connected to the SEIG for excitation purposes. The subsystem also includes a resistive primary load (7) and a main resistive load (4) connected to the generator. Similarly, the 5.5 kW SEIG subsystem comprises a prime mover (12) coupled with an induction generator (13), with its voltage regulated by another VFD (15). A capacitor (16) is used for excitation, and a resistive primary load (11) is connected to this subsystem. Both SEIG subsystems are connected in parallel through a synchronizing switch (8). This allows for the study of load sharing and stability between the two generators of different capacities. A Tektronix power analyser (9) is used to monitor the performance and electrical parameters of the parallel-operated SEIGs. To investigate the behaviour of the parallel-connected SEIGs under non-linear loading conditions, a rectifier (10) is included in the setup. The rectifier serves as a non-linear load, with an additional load (14) connected to its output. Furthermore, a main inductive load (17) is connected to the parallel-operated SEIGs to study the system's response to reactive power demands. Throughout the

experiments, a Fluke power analyser (2) is used to collect data and measure various electrical parameters, enabling a comprehensive analysis of the system's performance.

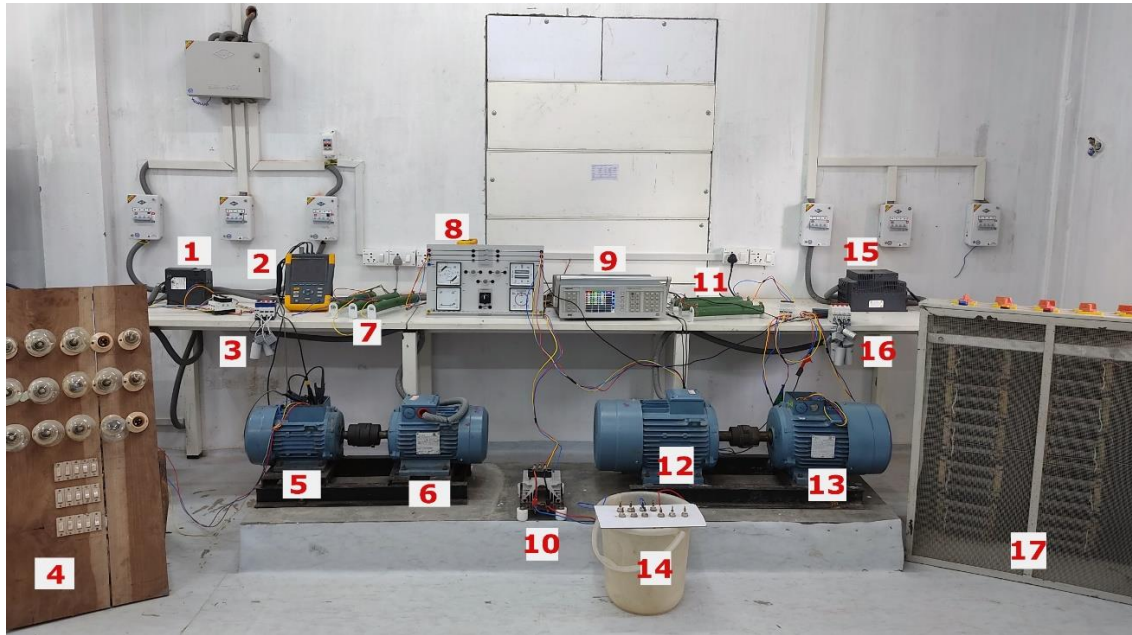


Fig. 5.2: Hardware set up of Parallel combination

5.5 Load sharing

The load connected to the parallel-connected SEIGs consists of a combination of resistive, inductive, and nonlinear components. The resistive load represents the electrical appliances and devices that consume active power, such as lighting and heating elements. The inductive load, such as motors and transformers, consumes reactive power and affects the system's power factor. The nonlinear load, such as power electronic converters and rectifiers, introduces harmonics into the system and can cause distortion in the voltage and current waveforms.

The load is connected to the SEIGs through a common bus, as shown in Figure 5.1. The bus voltage is maintained within the desired range by the combined action of the SEIGs and the excitation capacitor banks. The load sharing between the two SEIGs is determined by their respective power ratings and the excitation capacitance values.

$$S_1 = 2.2 \text{ kW}$$

$$S_2 = 5.5 \text{ kW}$$

$$\text{Load demand} = P_L + jQ_L$$

The total apparent power generated by the SEIGs.

$$S_T = S_1 + S_2$$

The power factor of the load ($\cos \phi_L$)

$$\cos \phi_L = \frac{P_L}{\sqrt{(P_L^2 + Q_L^2)}}$$

The total active and reactive power supplied by the SEIGs.

$$P_T = S_T \times \cos \phi_L$$

$$Q_T = S_T \times \sin \phi_L$$

The active and reactive power supplied by each SEIG based on their ratings.

$$P_1 = P_T \times \left(\frac{S_1}{S_T}\right)$$

$$Q_1 = Q_T \times \left(\frac{S_1}{S_T}\right)$$

$$P_2 = P_T \times \left(\frac{S_2}{S_T}\right)$$

$$Q_2 = Q_T \times \left(\frac{S_2}{S_T}\right)$$

The sum of the active and reactive powers supplied by the SEIGs equals the load demand.

$$P_L = P_T \times \left(\frac{S_1 + S_2}{S_T}\right)$$

$$Q_L = Q_T \times \left(\frac{S_1 + S_2}{S_T}\right)$$

Therefore, the active and reactive power supplied by the two SEIGs together matches the load demand, confirming that the load is shared between the SEIGs based on their power ratings.

The load sharing ratio are.

$$SEIG1: SEIG2 = S_1: S_2 = 2.2: 5.5 \approx 1: 2.5$$

This means that SEIG2 supplies approximately 2.5 times more power than SEIG1, which is consistent with their respective power ratings.

5.6 Advantages For parallel operation

Advantages of parallel operation of self-excited induction generators with a hybrid excitation system for rural electrification:

1. Increased power capacity: By paralleling multiple SEIG units, the total available power capacity can be increased to meet the varying electricity demands of rural communities. This scalability allows for gradual expansion as energy needs grow.

2. Improved reliability: With parallel operation, if one generator unit fails or requires maintenance, the others can continue supplying power, reducing the risk of complete system failure and ensuring continuous electricity supply.
3. Load sharing and flexibility: The parallel configuration allows for effective load sharing among the connected generators, providing flexibility to match the power output to varying load conditions. This is particularly useful in rural areas where load patterns can be unpredictable.
4. Efficient utilization of renewable resources: SEIGs can be driven by various renewable energy sources like wind, hydro, or biomass, which are often abundantly available in rural areas. Parallel operation enables the efficient harnessing of these distributed renewable resources.
5. Decentralized power generation: Parallel-connected SEIGs facilitate decentralized power generation, reducing the need for long-distance transmission lines and associated losses. This is advantageous for remote rural areas where grid extension may be impractical or costly.
6. Cost-effectiveness: Parallel operation allows for the use of smaller and more affordable SEIG units, which can be added incrementally as needed, reducing the initial capital investment required for rural electrification projects.
7. Modularity and scalability: The modular nature of parallel-connected SEIGs allows for easy expansion or reconfiguration as the energy needs of rural communities evolve, ensuring a flexible and adaptable power supply system.
8. Reduced environmental impact: By utilizing renewable energy sources and enabling decentralized power generation, parallel-connected SEIGs can contribute to reducing the environmental impact associated with conventional fossil fuel-based power generation.

CHAPTER 6

Results and discussion

CHAPTER 6

Results and discussion

The previous chapter focused on the examination and analysis of a hybrid excitation technique for parallel-connected self-excited induction generators (SEIGs). The study aimed to enhance the performance and efficiency of SEIGs in rural electrification applications by utilizing a combination of star and delta configurations for the capacitor banks, along with a neutral connection. The results demonstrated improved voltage regulation, load sharing, and efficiency under various load conditions. Building upon the findings of the previous chapter, this chapter presents a comprehensive study of the self-excited induction generator (SEIG) with a four-wire system, exploring its potential applications and operational characteristics. The research objectives outlined in this chapter are threefold:

Firstly, the aim to systematically analyse and evaluate various winding configurations for the SEIG, including delta and star connections for the machine windings, capacitor bank, and load connections. The purpose is to determine the most efficient and optimal combination for the four-wire system. By examining different winding configurations, the study seeks to identify the configuration that yields the best performance in terms of voltage regulation, power output, and overall system efficiency. Secondly the parallel operation of two micro-hydro power plants with capacities of 2.2 kW and 5.5 kW, employing a hybrid excitation technique. The hybrid excitation approach combines both star and delta configurations of the capacitor bank, along with a neutral connection. The objective is to assess the performance of the parallel-connected SEIGs in terms of load management, voltage regulation, and power quality. By analysing the behaviour of the system under various loading conditions and evaluating key performance indicators, the study aims to determine the effectiveness of the hybrid excitation technique in enhancing the reliability and stability of the parallel-operated SEIGs. Lastly a comprehensive understanding of the SEIG with a four-wire system, exploring its potential applications and operational characteristics. By conducting detailed experiments, simulations, and theoretical analyses, the study aims to establish the feasibility and benefits of employing SEIGs with a four-wire system in rural electrification projects. The research will investigate the system's ability to handle unbalanced loads, maintain stable voltage levels, and ensure reliable power supply to rural communities. The performance of parallel-operated SEIGs with hybrid excitation, and the overall potential of SEIGs in rural electrification applications. The findings will contribute to the advancement of SEIG technology and provide valuable insights for the design and implementation of efficient and reliable rural electrification systems.

6.1 Hybrid excitation approach

VOLTS/AMPS/HERTZ					POWER & ENERGY				
P _{UHI} 0:02:46					P _{UHI} 0:00:11				
	A	B	C	N		A	B	C	Total
V _{rms} _Δ	234.05	235.11	232.43	3.27	kW	- 0.723	- 0.726	- 0.711	- 2.159
	AB	BC	CA			A	B	C	Total
V _{rms} _Δ	404.24	406.54	404.37		kVA	0.723	0.726	0.711	2.159
	A	B	C	N		A	B	C	Total
A _{rms}	3.09	3.09	3.06	0.1	kvar	0.013	0.011	0.004	0.028
	A					A	B	C	Total
Hz	50.074				PF	-1.00	-1.00	-1.00	-1.00
17/11/23 15:40:58 239V 50Hz 3Ø WYE GR32144					17/11/23 15:41:59 239V 50Hz 3Ø WYE GR32144				
UP DOWN	TREND				UP DOWN	TREND			
	EVENTS 18					EVENTS 0			
	STOP START					STOP START			

Fig. 6.1: Hybrid excitation result (a) Volts, amps and Frequency (b) Power

The power and energy measurements shown in the image indicate that the SEIG system is operating in a regenerative mode, where power is being fed back into the system instead of being consumed by the load. This is evident from the negative active power (kW) values of -0.723, -0.726, and -0.711 for phases A, B, and C, respectively, resulting in a total active power of -2.159 kW. The negative sign signifies that the direction of power flow is reversed, with the SEIG acting as a source rather than a load, the power factor (PF) for all three phases is shown as -1.00, indicating that the current and voltage are 180 degrees out of phase. A power factor of -1.00 suggests that the SEIG is entirely supplying reactive power to the system, which is likely due to an overcompensation of reactive power by the hybrid capacitor bank configuration. The apparent power (kVA) values remain positive at 0.723, 0.726, and 0.711 for phases A, B, and C, respectively, with a total of 2.159 kVA. The apparent power represents the magnitude of the total power, regardless of the direction of power flow. The reactive power (kvar) values are relatively small, at 0.013, 0.011, and 0.004 for phases A, B, and C, respectively, with a total of 0.028 kvar. The regenerative operation of the SEIG in this case can be attributed to several factors. One possibility is that the hybrid excitation technique, which combines delta and star configurations for the capacitor banks, is providing excessive reactive power compensation. This overcompensation can lead to the SEIG generating more power than required by the load, resulting in the surplus power being fed back into the system. Another potential reason for the regenerative operation is that the SEIG may be driven above its synchronous speed. The hybrid excitation technique, coupled with the SEIG being driven above its synchronous speed, can contribute to the observed negative active power and power factor. For the performance of the SEIG system and to prevent undesired regenerative operation, it is essential to carefully select

and configure the capacitor banks used in the hybrid excitation technique. The capacitor values should be chosen based on the SEIG's parameters, load requirements, and desired operating conditions. Additionally, implementing a control system to regulate excitation and maintain stable operation can help mitigate the issues related to overcompensation and regenerative behaviour. the image indicates that the SEIG system with hybrid excitation is operating in a regenerative mode, with power being fed back into the system. This behaviour can be attributed to the overcompensation of reactive power by the hybrid capacitor configuration or the SEIG being driven above its synchronous speed. To achieve optimal performance and avoid regenerative operation, careful selection and configuration of the capacitor banks, along with appropriate control strategies, are necessary.

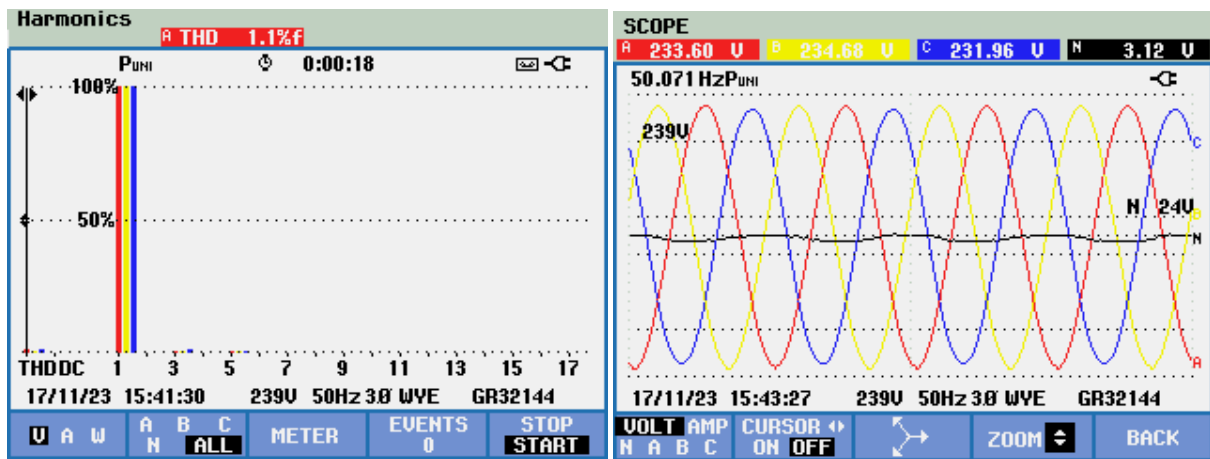


Fig. 6.2: Harmonics result and voltage waveform

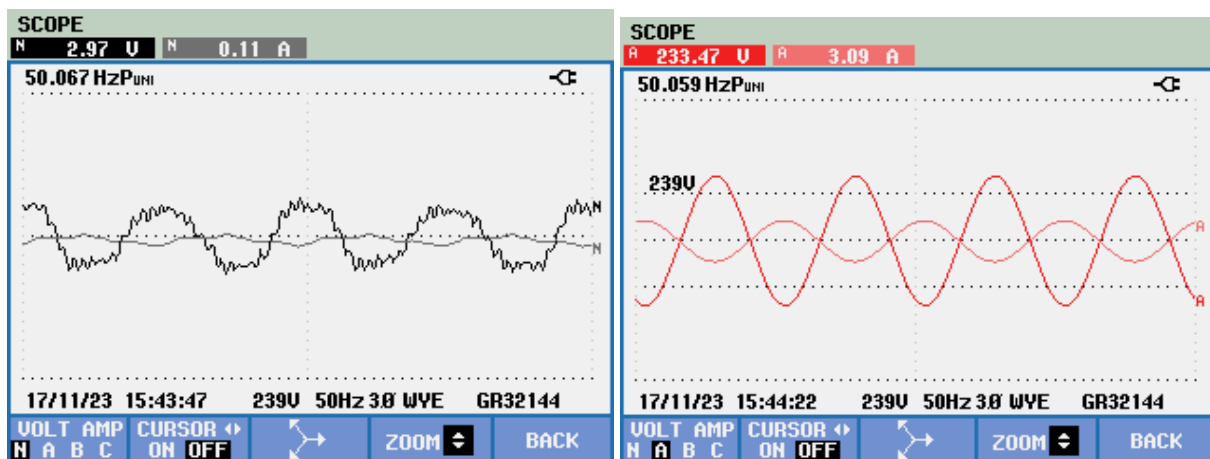


Fig. 6.3: (a) Neutral current and voltage waveform, (b) Single phase waveform

The 1.1% Total Harmonic Distortion (THD) in the excitation system as show in figure 6.2 (a) indicates excellent performance compared to traditional systems. Low THD means minimal harmonic content, resulting in reduced losses, less equipment stress, and improved power quality. The well-designed excitation system effectively mitigates harmonics, ensuring efficient

and reliable operation of the electrical system. The hybrid system's low neutral voltage (2.97V) and current (0.11A) as show in figure 6.3 (a) demonstrate its effectiveness in maintaining a balanced and stable neutral connection. By minimizing neutral voltage and current, the hybrid approach reduces the risk of overheating, equipment damage, and safety hazards associated with high neutral currents. The well-functioning neutral ensures proper phase balancing and enhances the overall performance and reliability of the electrical system.

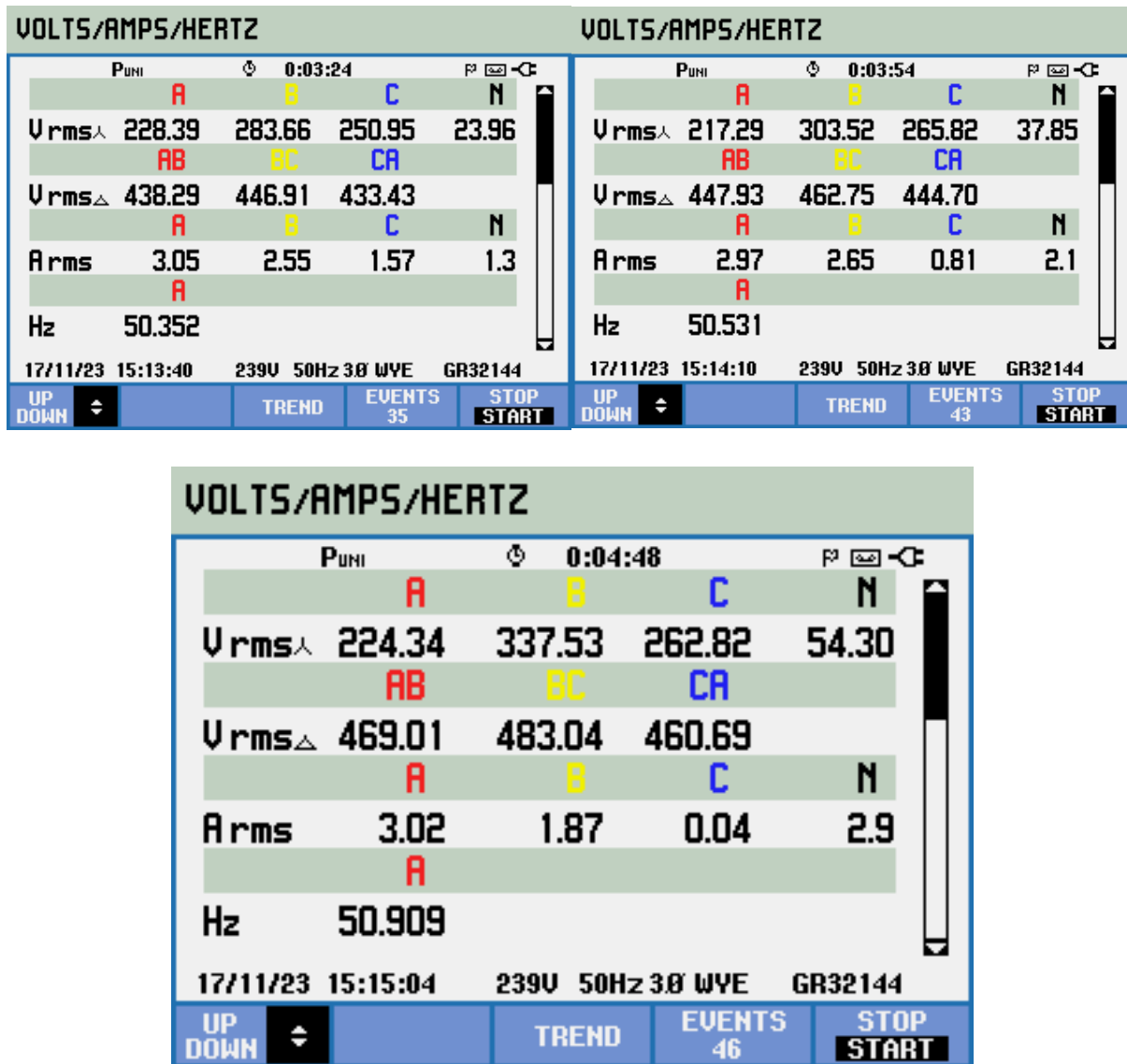


Fig. 6.4: Performance of a hybrid excitation system under three different unbalanced load condition

Case 1:

In this case, the R phase is subjected to a 100% load, the B phase to an 80% load, and the C phase to a 60% load. The voltages across the phases (V_{rms}) are relatively balanced, with values of 228.39V, 283.66V, and 250.95V for phases A, B, and C, respectively. The currents (I_{rms}) in the phases are 3.05A, 2.55A, and 1.57A, corresponding to the load percentages. The neutral

current is 1.3A, indicating some current flow in the neutral conductor due to the load imbalance. The frequency is stable at 50.352Hz.

Case 2:

In this case, the load distribution is changed, with the R phase at 80% load, the B phase at 70% load, and the C phase at only 20% load. The voltage values are 217.29V, 303.52V, and 265.82V for phases A, B, and C, respectively, showing a slight increase in voltage imbalance compared to case 1. The current values are 2.97A, 2.65A, and 0.81A, reflecting the reduced load on the C phase. The neutral current has increased to 2.1A due to the greater load imbalance. The frequency remains stable at 50.531Hz.

Case 3:

This case represents an extreme load imbalance, with the R phase at 100% load, the B phase at 50% load, and the C phase at 0% load (no load). The voltage values are 224.34V, 337.53V, and 262.82V for phases A, B, and C, respectively. The voltage imbalance is more pronounced, with the B phase voltage being significantly higher than the other phases. The current values are 3.02A, 1.87A, and 0.04A, with the C phase current being nearly zero due to the absence of load. The neutral current is 2.9A, the highest among the three scenarios, indicating a substantial current flow in the neutral conductor. The frequency remains stable at 50.909Hz.

The hybrid excitation system demonstrates its ability to maintain a relatively stable frequency across all three cases, despite the varying levels of load imbalance. However, as the load imbalance increases, the voltage imbalance becomes more evident, particularly in case 3, where the B phase voltage is noticeably higher than the other phases. The neutral current also increases with greater load imbalance, as seen in the progression from case 1 to case 3. This highlights the importance of properly sizing the neutral conductor to handle the expected neutral currents in unbalanced load conditions.

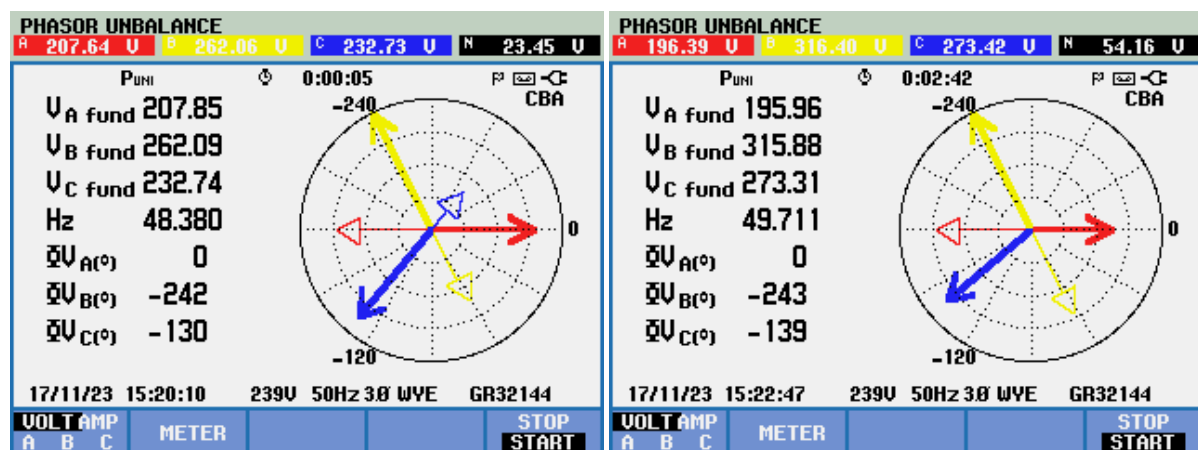


Fig. 6.5: Voltage phasor in two different condition

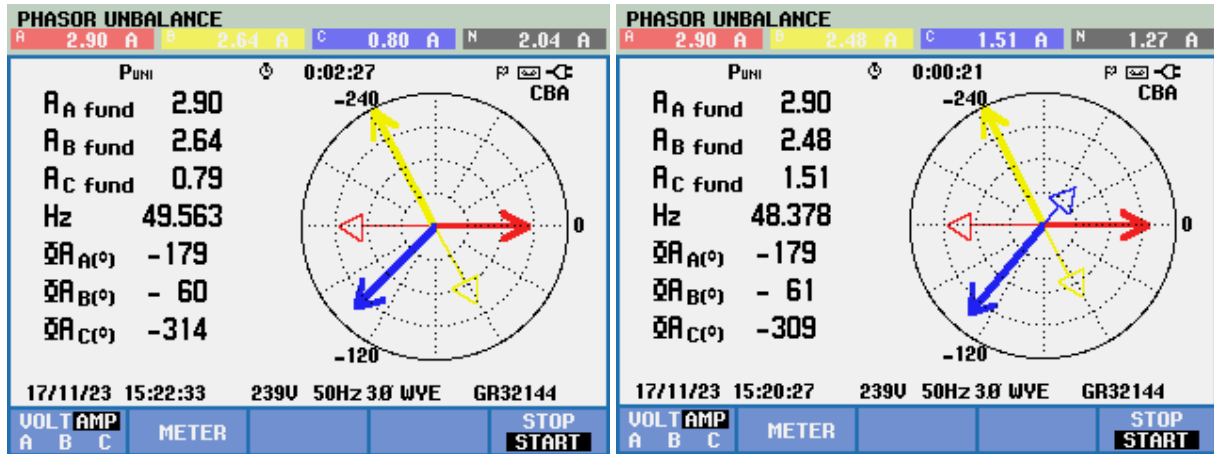


Fig. 6.6: Current phasor in two different condition

The phasor diagrams display the phase voltages (V_A , V_B , and V_C) and currents (I_A , I_B , and I_C) in a hybrid unbalanced condition, with the centre point representing the neutral or ground. In each figure, the phase voltages are not separated by 120 degrees, forming an unbalanced three-phase system. The phase currents are also not separated by 120 degrees,

the magnitudes of the phase voltages are not constant at around 240V, while the magnitudes of the phase currents gradually increase from approximately 0.7A to 1.5A. This suggests that the load is drawing more current over time while maintaining a constant voltage. The phasor diagrams also include a reference phasor (yellow arrow) at 0 degrees, which serves as a reference for measuring the phase angles of the voltages and currents. The phase angles remain not consistent across all four images, confirming that the system remains unbalanced and in phase throughout the observed period.

The hybrid excitation system validates its capability to operate under various unbalanced load conditions while maintaining a relatively stable frequency. The three cases tested showcase the system's response to different levels of load imbalance across the three phases. As the load imbalance increases from case 1 to case 3, the voltage imbalance becomes more pronounced, with the B phase voltage being significantly higher than the other phases in case 3. This voltage imbalance can have implications for the connected equipment and the overall power quality of the system. The neutral current also rises with increasing load imbalance, highlighting the importance of properly sizing the neutral conductor to accommodate the expected neutral currents. The highest neutral current is observed in case 3, where the load imbalance is most severe. Despite the challenges posed by unbalanced loads, the hybrid excitation system manages to maintain a stable frequency across all cases. This stability is crucial for the proper operation of connected equipment and the overall reliability of the electrical system.

The hybrid excitation system's ability to operate under unbalanced load conditions highlights its potential for applications where perfect load balancing may not always be feasible. However, further research and analysis can help refine the system's design and control strategies to better mitigate the effects of unbalanced loads and ensure optimal performance in various cases.

6.2 parallel operation of two SEIGs with hybrid excitation

6.2.1 R-Load

the parallel operation of two Self-Excited Induction Generators (SEIGs) with a hybrid excitation system under different load conditions. The system comprises two SEIGs with power ratings of 5.5 kW and 2.2 kW, which are subjected to resistive loads.

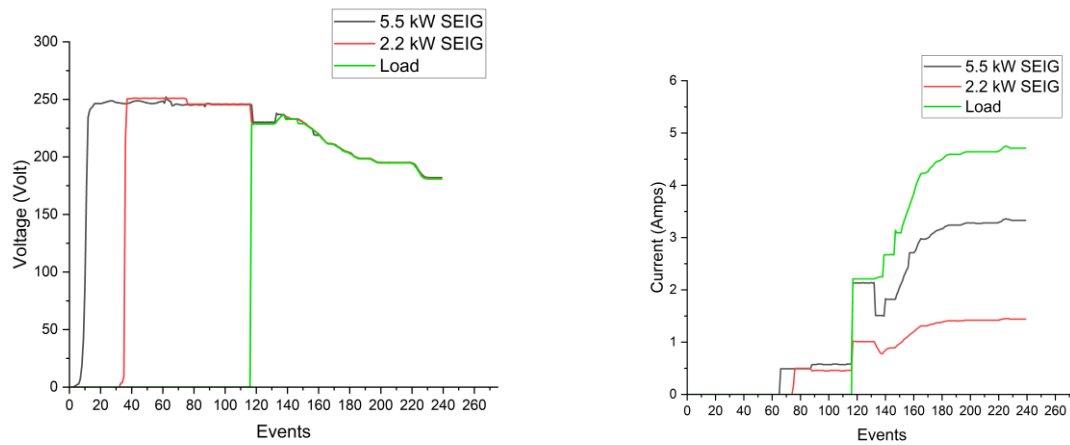


Fig. 6.7: Voltage and Current in R-load

The voltage vs. events graph (figure 6.7 (a)) showcases the system's behaviour during various events. The SEIGs start at events 6 and 34, respectively, and the voltage gradually builds up. At event 78, the primary loads of both SEIGs are turned on, and from event 84 to 117, the machines operate in parallel, maintaining a relatively stable voltage. The main load is introduced at event 117, causing a slight voltage dip. As the resistive load (R-load) gradually increases from event 138 onwards, the system voltage decreases accordingly, demonstrating the inverse relationship between load and voltage in a parallel SEIG system.

The current curve in Figure 6.7(b) reveals that the 5.5 kW SEIG shares more current compared to the 2.2 kW SEIG when the machines start operating in parallel mode at event 84. This indicates that the load sharing between the two SEIGs is not equal, with the larger SEIG contributing more to the overall current supply.

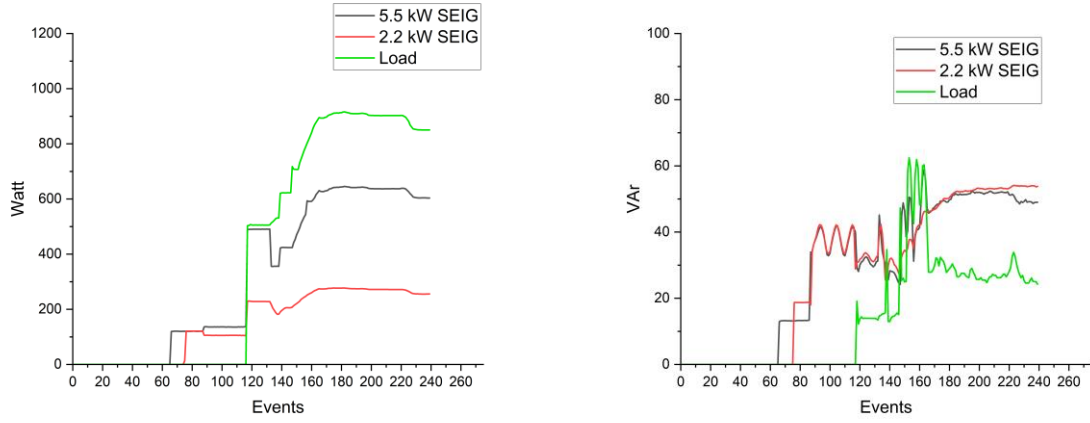


Fig. 6.8: Active and Reactive Power in R-load

Figure 6.8(a) shows that the R-load consumes 900 W of active power, with the 5.5 kW SEIG contributing 630 W and the 2.2 kW SEIG contributing 270 W. This unequal active power sharing is due to the difference in the power ratings of the two SEIGs. The reactive power consumption of the SEIGs, however, is not always proportional to their power ratings and can vary based on factors such as their loading and position on the magnetization curve in figure 6.8 (b). The hybrid excitation system plays a crucial role in providing extra reactive power for both the load and the SEIGs to maintain stability. The SEIGs' reactive power consumption can change based on their operating conditions, and the hybrid capacitors in star and delta configurations help provide the needed reactive power, adjusting to the requirements of the SEIGs and the load.

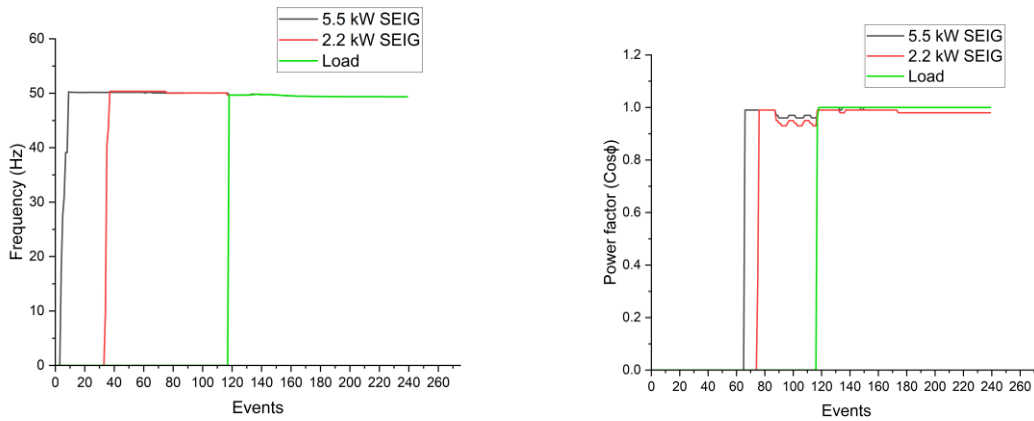


Fig. 6.9: frequency and Power factor in R-load

Figure 6.9(a) confirms that both SEIGs operate at the same frequency after the parallel connection, indicating a successful synchronization process. This is crucial for the stable and synchronized operation of the hybrid system. The hybrid capacitors are designed to compensate

for the reactive power requirements of the SEIGs during the self-excitation process. Once the machines are synchronized and connected to a resistive load, the capacitors effectively fulfil the reactive power needs of the generators, allowing them to operate at close to the unity power factor, as shown in figure 6.9(b). This near-unity power factor operation ensures efficient power transfer and minimizes losses in the system.

6.2.2 R with 2L Load

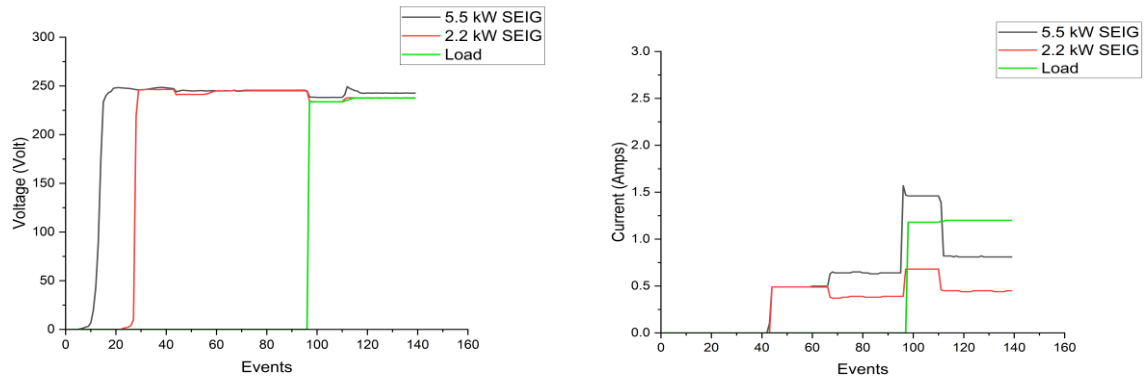


Fig. 6.10: Voltage and Current in R-2L load

The behaviour of a parallel operation of two Self-Excited Induction Generators (SEIGs) with a hybrid excitation system when connected to a resistive load in series with a 2 mH inductive load (R-2 mH L Load). The system comprises two SEIGs with power ratings of 5.5 kW and 2.2 kW. The voltage vs. events graph shows that the SEIGs start at events 6 and 34, respectively, and the voltage gradually builds up as the machines begin to generate power. The voltage continues to rise and stabilizes as the SEIGs operate independently until event 78, when the primary loads of both SEIGs are turned on, preparing the system for parallel operation.

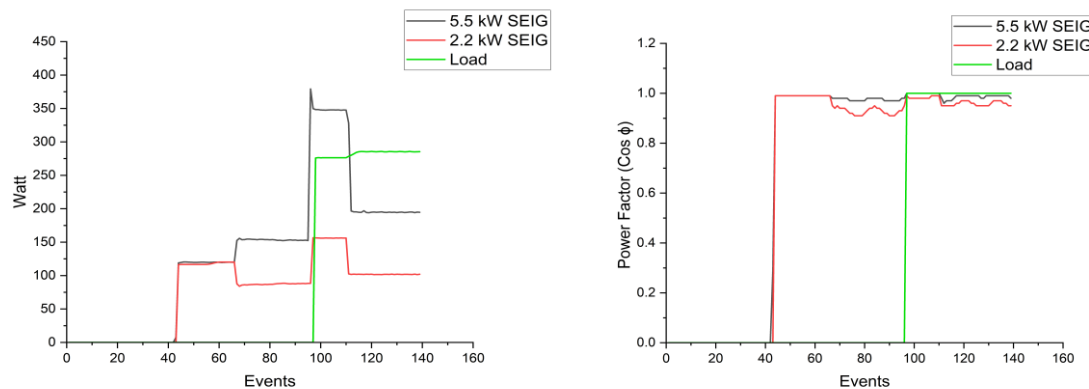


Fig. 6.11: Active power and Power Factor in R-2L Load

From events 84 to 117, the machines operate in parallel, maintaining a relatively stable voltage. The current vs. events graph reveals that the 5.5 kW SEIG contributes more current than the 2.2 kW SEIG during parallel operation due to their different power ratings. At event 117, the main R-2mH L Load is introduced, causing a slight voltage dip. The active power consumed by the load is shared between the SEIGs, with the 5.5 kW SEIG contributing more than the 2.2 kW SEIG, as shown in the active power graph.

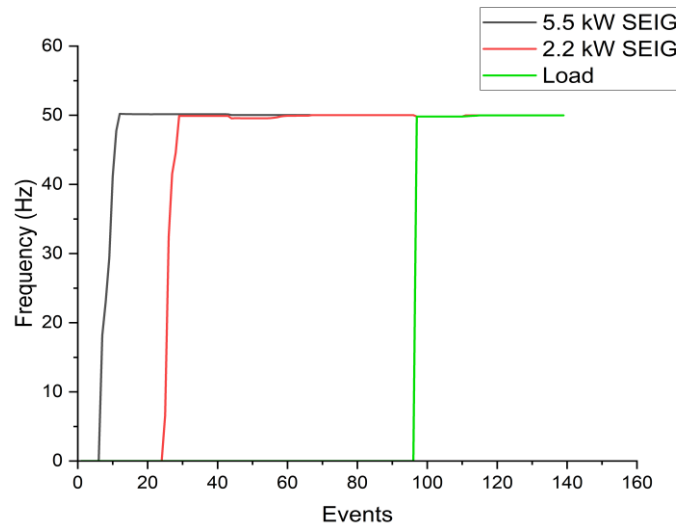


Fig. 6.11: Frequency in R-2L Load

As the R-2mH L Load gradually increases from events 138 to 160, the system voltage decreases, similar to the behaviour observed with the purely resistive load. However, the overall voltage level is slightly lower due to the inductive component of the load. The power factor vs. events graph indicates that the power factor of the system is lower compared to the purely resistive load case. This is because the inductive component (2 mH) of the load consumes reactive power, reducing the overall power factor. The hybrid excitation system, including the capacitors, works to compensate for the increased reactive power requirements of the SEIGs and the load to maintain stability and improve the power factor.

Throughout the events, the parallel operation of the SEIGs remains stable, with both machines operating at the same frequency. The hybrid excitation system plays a crucial role in maintaining the system's stability by supplying the required reactive power to the SEIGs and the load. The presence of the inductive component in the R-2mH L Load affects the system's voltage, current, active power, and power factor. The voltage drop is similar to the purely resistive load case but with a slightly lower overall voltage level. The power factor is reduced due to the increased reactive power consumption of the inductive load.

6.2.3 R with 5L Load

The load connected to the system is a combination of a resistive load and a 5 mH inductive load in series (R-5mH L Load). The voltage graph shows that the SEIGs start generating power at events 6 and 34, respectively, and the voltage gradually builds up. As the machines continue to operate independently, the voltage stabilizes until event 78, when the primary loads of both SEIGs are turned on in preparation for parallel operation.

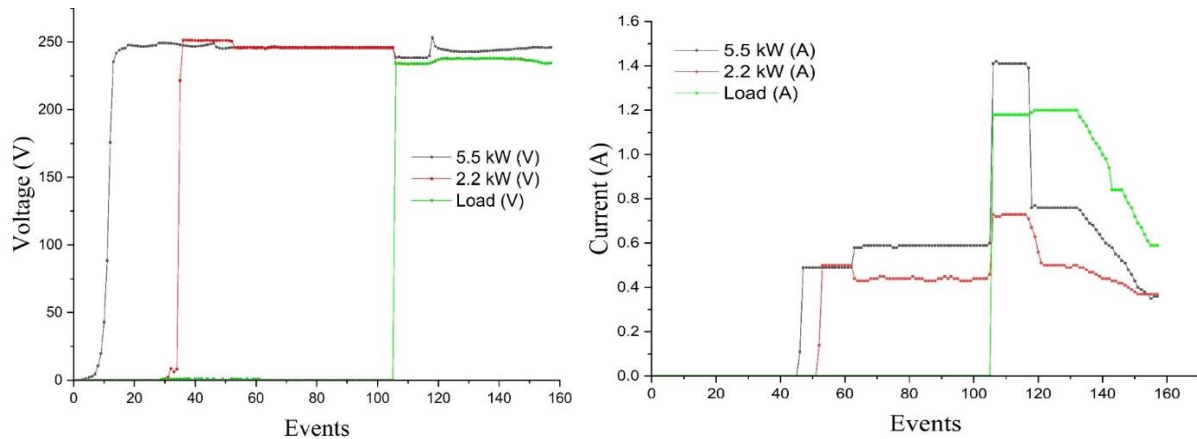


Fig. 6.12: Voltage and Current in R-5L load

From events 84 to 117, the SEIGs operate in parallel, maintaining a relatively stable voltage. However, the current graph reveals that the 5.5 kW SEIG contributes more to the total current compared to the 2.2 kW SEIG due to their difference in power ratings.

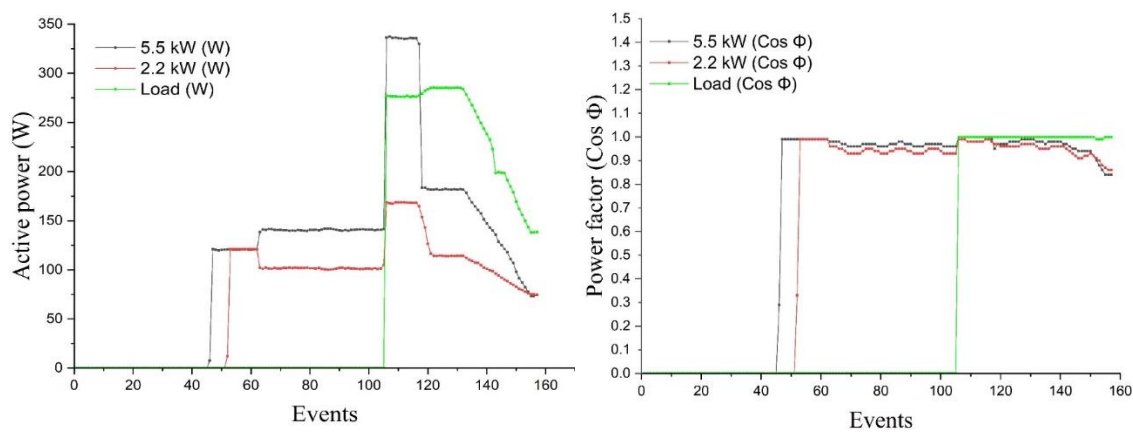


Fig. 6.13: Active power and Power Factor in R-5L Load

At event 117, the main R-5mH L Load is introduced, causing a significant voltage dip, which is more pronounced than in the previous cases with purely resistive and R-2mH L Loads. The active power graph shows that the load consumes power, with the 5.5 kW SEIG contributing more than the 2.2 kW SEIG.

As the R-5mH L Load increases from events 138 to 160, the system voltage experiences a substantial drop, which is more severe compared to the previous load cases. This is attributed to the higher inductive component (5 mH) of the load, which consumes more reactive power and leads to a greater voltage drop.

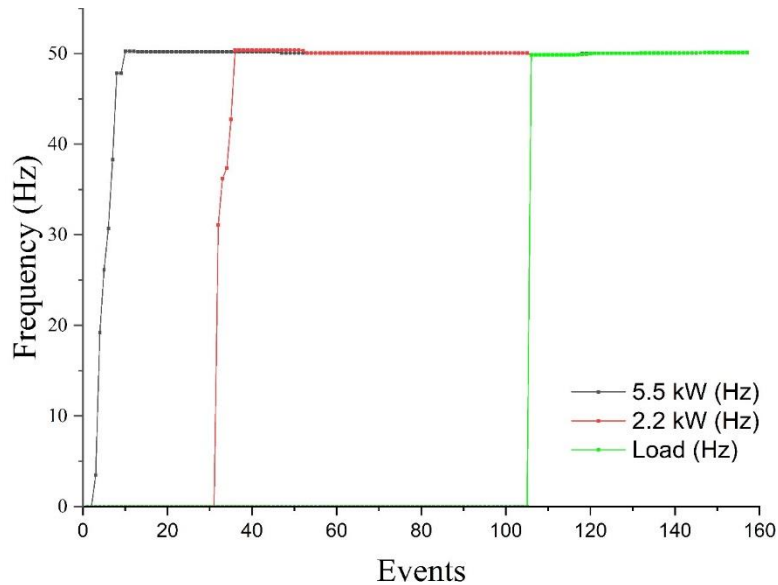


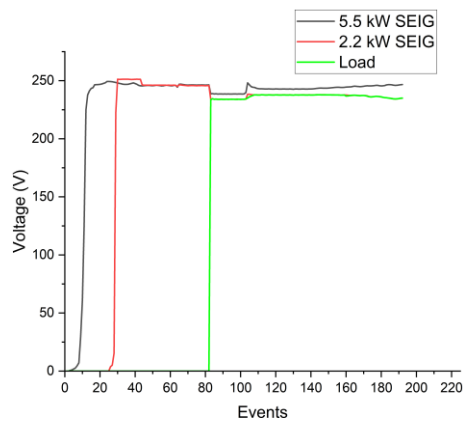
Fig. 6.14: Frequency in R-5L Load

The power factor graph indicates that the overall power factor of the system is significantly lower compared to both the purely resistive and R-2mH L Load cases. The increased inductive component (5 mH) of the load consumes more reactive power, resulting in a further reduction of the power factor. Consequently, the hybrid excitation system, including the capacitors, has to work harder to compensate for the higher reactive power demand of the SEIGs and the load to maintain stability and improve the power factor. Despite the challenging load conditions, the parallel operation of the SEIGs remains stable throughout the experiment, with both machines operating at the same frequency. The hybrid excitation system plays a crucial role in maintaining the system's stability by providing the necessary reactive power to the SEIGs and the load.

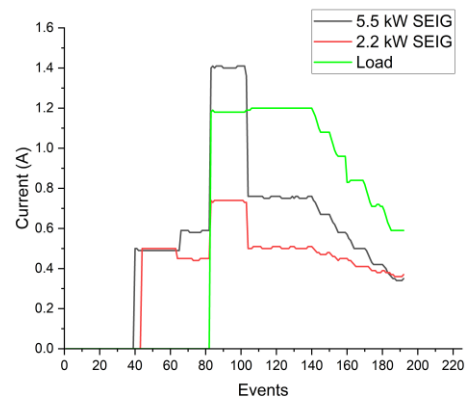
6.2.4 R with 7L Load

The experimental results for the resistive load in series with a 7 mH inductive load (R-7mH L Load) closely resemble those of the R-5mH L Load case. The voltage, current, active power, and power factor graphs exhibit similar patterns and trends, with only minor differences between the two cases. This suggests that increasing the inductive component from 5 mH to 7 mH does not significantly alter the system's behaviour or the challenges faced by the hybrid

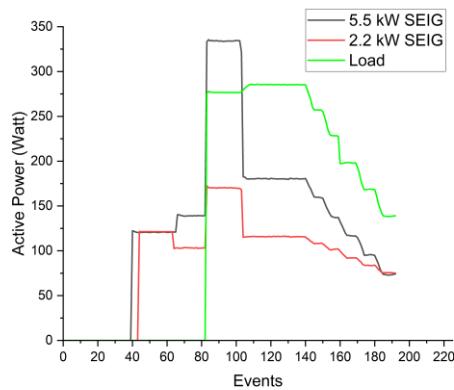
excitation system in maintaining stability and compensating for the reactive power requirements of the SEIGs and the load under these specific operating conditions.



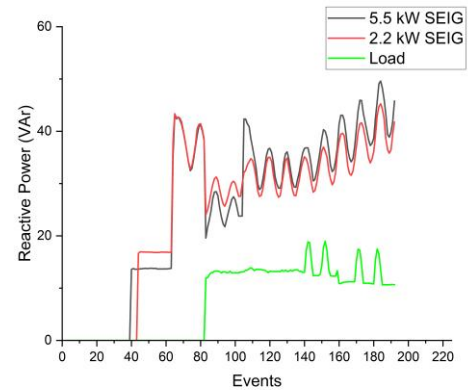
(a)



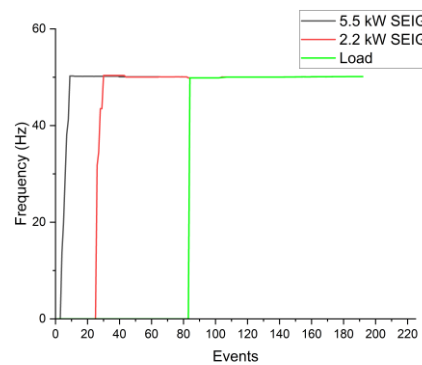
(b)



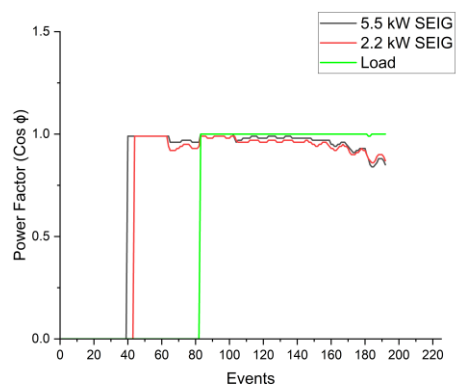
(c)



(d)



(e)



(f)

Fig. 6.15: (a) Voltage, (b) Current, (c) Active Power, (d) reactive power, (e) Frequency and (f) Power Factor

6.2.5 Non-linear Load

In this experimental setup, the load connected to the parallel-operated Self-Excited Induction Generators (SEIGs) with a hybrid excitation system is a non-linear load.

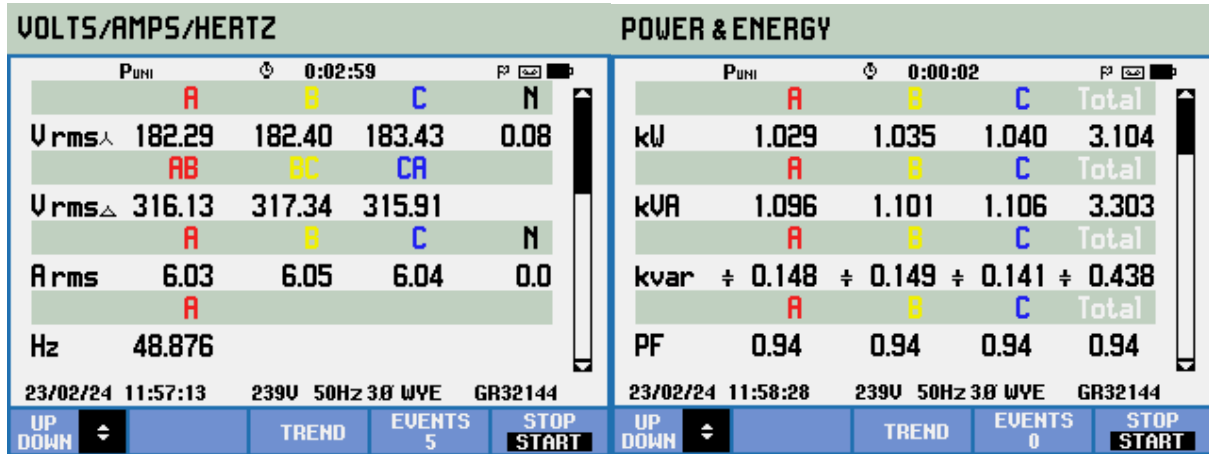


Fig. 6.16: (a) Voltage, Current, Active Power, reactive power, (b) Frequency and Power Factor

The non-linear load consists of a rectifier with a 1 kW heater connected to it. Non-linear loads, such as rectifiers, introduce harmonics and distortions in the system, which can affect the voltage and current waveforms, power quality, and the overall performance of the SEIGs and the hybrid excitation system. When the non-linear load is connected to the system, the voltage graph may show distortions and deviations from the ideal sinusoidal waveform. The presence of harmonics can cause the voltage to have a non-sinusoidal shape, leading to a decrease in the overall power quality. The current graph may also exhibit distortions due to the non-linear nature of the load. The rectifier draws current in a non-sinusoidal manner, which can result in a higher peak current compared to the RMS (root mean square) value.

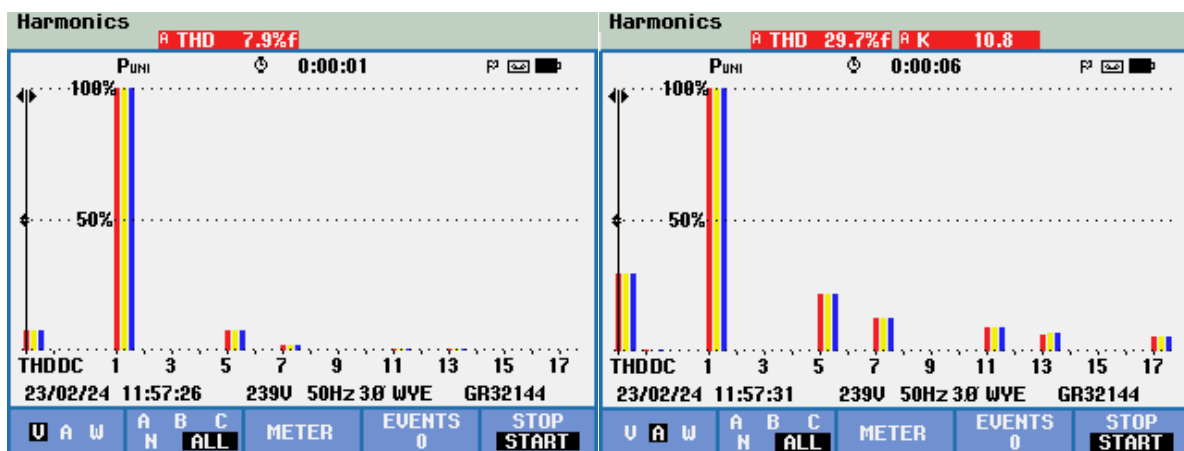


Fig. 6.17: Voltage and current THD

This non-sinusoidal current can cause additional heating in the SEIGs and the system components, potentially leading to reduced efficiency and increased losses. The active power consumed by the non-linear load (1 kW heater) may be affected by the presence of harmonics and distortions.

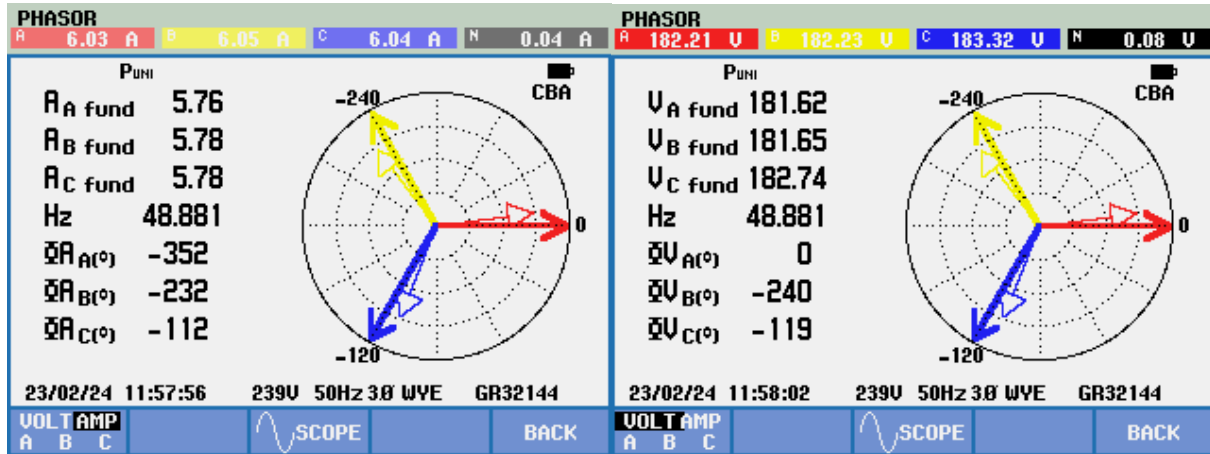


Fig. 6.17: Voltage and current Phasor

The true power consumed by the load may differ from the apparent power due to the non-sinusoidal nature of the voltage and current waveforms. The power factor of the system with the non-linear load may be lower compared to the previous cases with linear loads (resistive and resistive-inductive). The rectifier introduces a reactive power component, which can further decrease the power factor. The hybrid excitation system, including the capacitors, will work to compensate for the reactive power requirements and maintain the stability of the parallel SEIG operation under these non-linear load conditions.

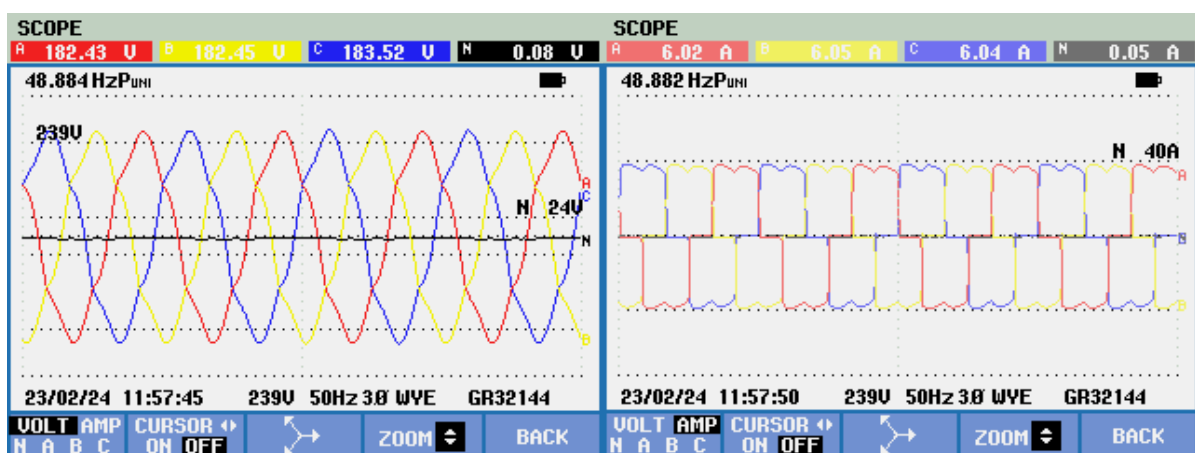


Fig. 6.17: Voltage and current waveform

The harmonics and distortions introduced by the non-linear load can also affect the synchronization and stability of the parallel-operated SEIGs. The hybrid excitation system plays a vital role in maintaining the system's stability by providing the necessary reactive power compensation and minimizing the impact of harmonics on the SEIGs' performance.

The performance of a parallel operation of two Self-Excited Induction Generators (SEIGs) with a hybrid excitation system when connected to a non-linear load, consisting of a rectifier with a 1 kW heater. The voltage graph shows that the SEIGs start generating power at events 6 and 34, respectively, and the voltage rises and stabilizes as the machines operate independently until event 78, when the primary loads are turned on for parallel operation.

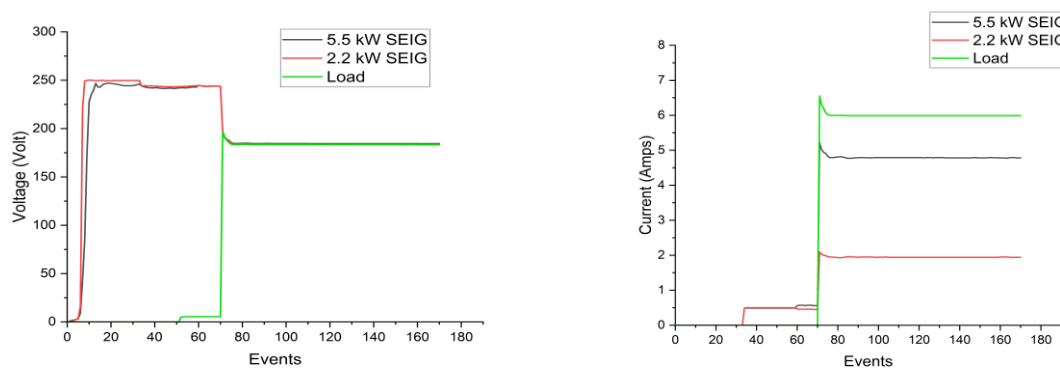


Fig. 6.18: Voltage and current in non-linear load

From events 84 to 117, the SEIGs operate in parallel, but the voltage graph displays distortions and deviations from the ideal sinusoidal waveform due to the harmonics introduced by the non-linear load. The current graph also exhibits distortions, with the 5.5 kW SEIG contributing more current than the 2.2 kW SEIG during parallel operation. At event 117, the main non-linear load is introduced, causing further distortions in the voltage and current waveforms. The active power consumed by the load is shared between the SEIGs, with the 5.5 kW SEIG contributing more than the 2.2 kW SEIG.

As the non-linear load continues to operate from events 138 to 180, the voltage and current distortions persist, with a decrease in the overall voltage level compared to the previous linear load cases. The power factor of the system with the non-linear load is significantly lower than in the linear load cases, as the rectifier introduces a reactive power component, further decreasing the power factor. The VAR graph shows a higher reactive power demand compared to the linear load cases.

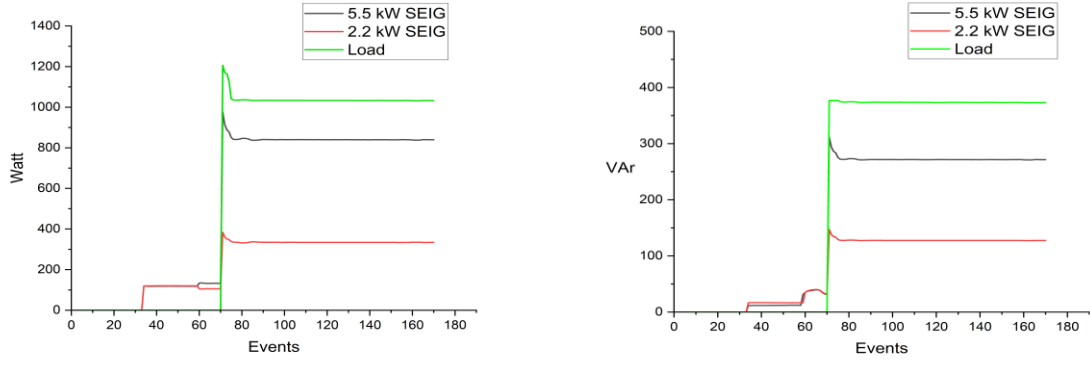


Fig. 6.19: Active and Reactive Power in non-linear load

The hybrid excitation system, including the capacitors, works to provide the necessary reactive power compensation to maintain the system's stability and minimize the impact of harmonics on the SEIGs' performance. Despite the challenges introduced by the non-linear load, the parallel operation of the SEIGs remains stable throughout the events, highlighting the effectiveness of the hybrid excitation system in maintaining the system's stability under these demanding conditions.

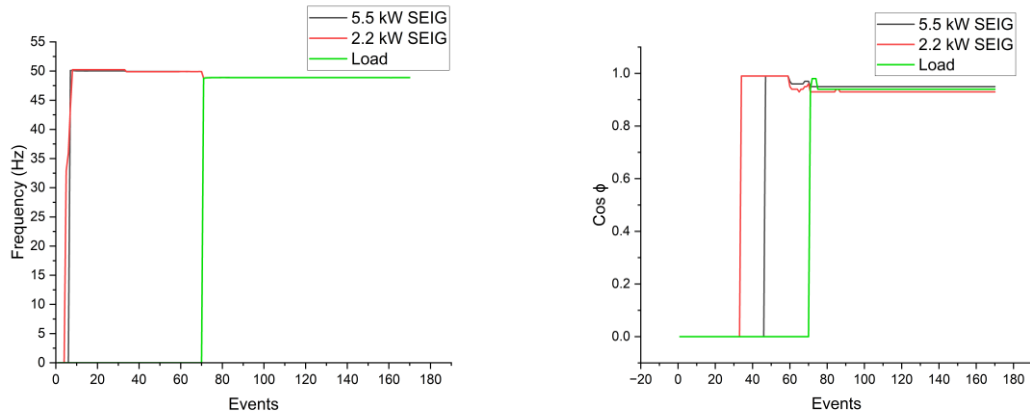


Fig. 6.20: Frequency and Power Factor in non-linear load

In the non-linear load significantly impacts the system's performance, causing distortions in the voltage and current waveforms, reducing the power factor, and increasing the reactive power demand. The hybrid excitation system plays a vital role in compensating for the reactive power requirements and maintaining the stability of the parallel SEIG operation, ensuring the system remains stable despite the challenges posed by the non-linear load.

The results demonstrate the effectiveness of the hybrid excitation system in maintaining stability, power quality, and efficient operation of the parallel-connected SEIGs. Under resistive load conditions, the system maintains a stable voltage and frequency, with the SEIGs sharing

the load according to their power ratings. The hybrid excitation system effectively compensates for the reactive power requirements of the SEIGs and the load, ensuring a near-unity power factor operation. As the load increases, the system voltage decreases, highlighting the inverse relationship between load and voltage in a parallel SEIG system. The introduction of inductive loads, such as the R-2mH L Load and R-5mH L Load, presents additional challenges to the system. The inductive component of the load consumes reactive power, leading to a more pronounced voltage drop and a lower power factor compared to the purely resistive load case. The hybrid excitation system, including the capacitors, works harder to compensate for the increased reactive power demand and maintain the system's stability. The results show that the system can handle inductive loads effectively, with the hybrid excitation system adapting to the changing reactive power requirements. The non-linear load, consisting of a rectifier with a 1 kW heater, introduces significant distortions in the voltage and current waveforms, reduces the power factor, and increases the reactive power demand. The hybrid excitation system plays a vital role in compensating for the reactive power requirements and minimizing the impact of harmonics on the SEIGs' performance. Despite the challenges posed by the non-linear load, the system remains stable, highlighting the robustness and effectiveness of the hybrid excitation system in maintaining the parallel operation of SEIGs under demanding conditions. The successful parallel operation of SEIGs with a hybrid excitation system under various load conditions demonstrates its potential for applications in distributed generation and renewable energy systems. The hybrid excitation system's ability to maintain stability, power quality, and efficient operation, even in the presence of challenging loads, makes it a promising solution for the integration of renewable energy sources into the power grid.

CHAPTER 7

Conclusion

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Conclusion

The comprehensive study on the hybrid excitation system for parallel-connected self-excited induction generators (SEIGs) has yielded valuable insights into its performance, stability, and potential applications in distributed generation and renewable energy systems. The research encompassed a thorough analysis of the system's behaviour under various load conditions, including balanced resistive loads, unbalanced loads, inductive loads, and non-linear loads.

The experimental results have demonstrated the effectiveness of the hybrid excitation system in maintaining the stability and power quality of the parallel-connected SEIGs. Under balanced resistive load conditions, the system exhibited stable voltage and frequency characteristics, with the SEIGs sharing the load proportionally to their power ratings. The hybrid excitation system, comprising carefully designed capacitor banks, successfully compensated for the reactive power requirements of the SEIGs and the load, ensuring near-unity power factor operation. This highlights the system's ability to optimize power flow and minimize losses in the distribution network. The investigation of the hybrid excitation system's performance under unbalanced load conditions revealed its robustness and adaptability. As the load imbalance increased, the system experienced voltage imbalances and increased neutral currents. However, the hybrid excitation system demonstrated its capability to maintain a relatively stable frequency despite these challenges. The findings underscore the importance of proper system design, including the sizing of neutral conductors and the implementation of control strategies to mitigate the effects of unbalanced loads on power quality and system stability. The introduction of inductive loads, such as the R-2mH L Load and R-5mH L Load, posed additional challenges to the hybrid excitation system. The inductive component of the load consumed reactive power, leading to more pronounced voltage drops and lower power factors compared to purely resistive loads. The hybrid excitation system, including the capacitor banks, adapted to the changing reactive power requirements and worked to maintain the system's stability. The results confirmed the system's ability to handle inductive loads effectively, ensuring the reliable operation of the parallel-connected SEIGs.

The study also explored the impact of non-linear loads on the hybrid excitation system, using a rectifier with a 1 kW heater as a representative example. The non-linear load introduced significant distortions in the voltage and current waveforms, reduced the power factor, and increased the reactive power demand. The hybrid excitation system played a crucial role in

compensating for the reactive power requirements and minimizing the impact of harmonics on the SEIGs' performance. Despite the challenges posed by the non-linear load, the system remained stable, demonstrating the robustness and effectiveness of the hybrid excitation system in maintaining the parallel operation of SEIGs under demanding conditions.

The successful parallel operation of SEIGs with a hybrid excitation system under various load conditions highlights its potential for applications in distributed generation and renewable energy systems. The hybrid excitation system's ability to maintain stability, power quality, and efficient operation, even in the presence of challenging loads, makes it a promising solution for the integration of renewable energy sources into the power grid. The system's adaptability and robustness can facilitate the deployment of SEIGs in remote and off-grid locations, providing reliable and sustainable electricity to communities. The hybrid excitation system for parallel-connected SEIGs has demonstrated its effectiveness in maintaining stability, power quality, and efficient operation under various load conditions. The system's adaptability and robustness make it a promising solution for the integration of renewable energy sources into the power grid and the deployment of SEIGs in distributed generation applications. With further research and optimization, the hybrid excitation system can play a vital role in the transition towards sustainable and reliable energy systems.

CHAPTER 8

Future Work

CHAPTER 8

Future Work

1. Advanced Control Strategies:

Develop and implement advanced control strategies for the hybrid excitation system to enhance its performance and adaptability. Investigate techniques such as intelligent load management, adaptive excitation control, and predictive control algorithms to optimize the system's response to varying load conditions and improve voltage regulation. These control strategies can leverage real-time data acquisition and machine learning algorithms to make informed decisions and ensure optimal system operation.

2. Grid Integration of Parallel-Connected SEIGs:

Explore the integration of parallel-connected SEIGs with the utility grid using the hybrid excitation system. Investigate the challenges and opportunities associated with grid synchronization, power flow control, and fault ride-through capabilities. Develop robust control schemes and protection systems to ensure seamless integration and stable operation of the SEIGs in grid-connected mode. Analyze the impact of the hybrid excitation system on grid stability, power quality, and the overall performance of the integrated system.

3. Unbalanced Load Mitigation Techniques:

Further investigate techniques to mitigate the effects of unbalanced loads on the hybrid excitation system and the parallel-connected SEIGs. Explore advanced load balancing strategies, such as dynamic load redistribution and phase-shifting transformers, to minimize voltage imbalances and reduce neutral currents. Develop intelligent algorithms to detect and compensate for load imbalances in real-time, ensuring stable and efficient operation of the system under various unbalanced load scenarios.

4. Harmonic Mitigation and Power Quality Improvement:

Conduct in-depth studies on the impact of non-linear loads and harmonics on the hybrid excitation system and the parallel-connected SEIGs. Develop effective harmonic mitigation techniques, such as active power filters and harmonic compensation algorithms, to reduce the distortions in voltage and current waveforms and improve power quality. Investigate the

potential of integrating advanced power electronic converters and control schemes to enhance the system's ability to handle non-linear loads and maintain a high-quality power supply.

5. Scalability and Optimization of the Hybrid Excitation System:

Investigate the scalability of the hybrid excitation system for larger-scale applications and higher power ratings. Explore the optimal sizing and configuration of the capacitor banks, considering factors such as system capacity, load characteristics, and economic constraints. Develop design guidelines and optimization tools to assist in the planning and deployment of hybrid excitation systems for various distributed generation and renewable energy applications.

6. Techno-Economic Analysis and Life Cycle Assessment:

Conduct comprehensive techno-economic analyses to evaluate the feasibility and viability of implementing the hybrid excitation system in different application scenarios. Assess the initial investment costs, operating expenses, and potential energy savings associated with the system. Perform life cycle assessments to quantify the environmental impacts and sustainability benefits of the hybrid excitation system compared to conventional excitation methods. Identify potential barriers to adoption and develop strategies to overcome them, considering technical, economic, and policy perspectives.

Appendix 1

TABLE A2 MACHINE PARAMETERS.

<i>Sl.no.</i>	<i>Parameters</i>	<i>SEIG</i>
1	<i>Ratings</i>	2.2 kW
2	<i>Current</i>	4.8 A
3	<i>Power Factor (Cos ϕ)</i>	0.76
4	<i>Voltage</i>	415 V

Appendix 2

TABLE A2 EQUIVALENT CIRCUIT PARAMETERS.

<i>SL.NO.</i>	<i>PARAMETERS</i>	<i>SEIG</i>
1	R_r	2.87 Ω
2	R_s	3.81 Ω
3	$X_{ls} = X_{lr}$	4.45 Ω
4	X_m	87.010 Ω
5	<i>Base impedance 'e'</i>	86.113
6	R_m	105.24 Ω
7	R_L	500 Ω
8	C_{Δ}	12.5 μF
9	C_Y	45 μF
10	$C_{\Delta Y}$	12.5 μF & 2 μF

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